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20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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1 Abstract

This thesis examines how the parliamentary representation of smaller parties is affected by the transition from Hagenbach-Bischoff to Double Pukelsheim. To do so, all 31 available cantonal parliamentary elections for the eight transition cantons between 2007 and 2026 are analyzed. The smaller parties groups, combines all electoral lists and parties which received less than 10% of the total votes in the analyzed area. To make this comparison, the offical results under Double Pukelsheim are taken and the counterfactual seat allocation under Hagenbach-Bischoff is computed. This while including the former canton electoral law and holding electoral units, observed votes, seat totals and reported list structures constant.

The main finding is that across all 31 elections, the smaller parties aggregately receive a 7.6 percentage points higher seat-share under Double Pukelsheim than under the conterfactual Hagenbach-Bischoff. An aggregate gain for the smaller parties occurs in 30 of 31 elections. Further is the observed gain persistent to alternative smaller-party thresholds, across succesive elections and to the omission of individual cantons. While analyses regarding the pre-reform district magnitude or explicit electoral thresholds, reveal no consistent relationship, respectively are sensitive to the inclusion of individual cantons.

The persistent mechanical seat-share gain for smaller parties observed here, indicate the reduction of the prior representational disadvantage under the Double Pukelsheim system. This while behavioural responses to the reform remain outside the scope of the analysis.

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2 List of Abbreviations

AG	Canton of Aargau
ALG	Alternative – the Greens Zug (Alternative – die Grünen Zug)
BDP	Conservative Democratic Party
BGE	Decisions of the Swiss Federal Supreme Court (official collection)
CSV	Comma-separated values
CV	Coefficient of variation
CVP	Christian Democratic People's Party
DP	Double Pukelsheim system
EQ1	Exploratory question 1 (district magnitude)
FDP	FDP.The Liberals
FSC	Swiss Federal Supreme Court
GLP	Green Liberal Party
GR	Canton of Graubünden
HB	Hagenbach-Bischoff method
JFSH	Young Liberals Schaffhausen (Jungfreisinnige Schaffhausen)
JUSO	Young Socialists Switzerland
NW	Canton of Nidwalden
PDF	Portable Document Format
pp	percentage points
PR	proportional representation
Q1	First quartile (25th percentile)
Q3	Third quartile (75th percentile)
SH	Canton of Schaffhausen
SP	Social Democratic Party of Switzerland
SVP	Swiss People's Party
SZ	Canton of Schwyz
UR	Canton of Uri
VS	Canton of Valais
XLSX	Excel workbook file format (Office Open XML)
ZG	Canton of Zug
ZH	Canton of Zurich

3 Introduction

On the 18 December 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 I 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds for parties to win a seat varied heavily from district to district. Further did some districts have thresholds for parties to win a seat of up to 34%. Consequently, votes did not have equal effect, which undermines the electoral equality included in Art. 8 (Equality before the law) and Art. 34 (political rights) of the Federal Constitution. Based on this criticism the court decided that the City of Zurich had to implement corrective measures, but could choose their form (BGE 129 I 185 E. 7.3, E. 7.6.2–7.6.3 and E. 8.3.2; Schweizerische Eidgenossenschaft, 1999, Arts. 8 and 34). The Government of the Canton of Zurich decided to keep the constituency structure and instead commissioned the german mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a method called "biproportional divisor method with standard rounding" which is today often simply called "Double Pukelsheimer" or "Biproportional apportionment". (Regierungsrat des Kantons Zürich, 2014, pp. 3–4; Lüscher, 2024, pp. 9-10)

The distinguishing feature of the Pukelsheimer Proporz to other proportional representation systems, is the so called upper-apportionment. It artificially creates a single electoral district for the initial allocation of the parliamentary seats to the parties or list groups. Through this mechanism, the difference between seat and voteshares gets minimized and with that also the effective threshold for parties to enter the parliament. (Lüscher, 2023, p. 575)

Through its upper-apportionment mechanism and the lowered threshold, the reform is especially relevant for the representation of smaller parties, since votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level. Sandro Lüscher illustrates this mechanical effect with the case of Schwyz's Landrat election in 2016, there under the old Hagenbach-Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they received three seats (Lüscher, 2025, p. 1-3).

Since Zurich first used the system in 2007, eight further cantons have used it: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016) , Valais (2017) , Uri (2020), Graubünden (2022). In 2025 also the voters from Basel Landschaft decided to implement the biproportional apportionment from 2027 on (Lüscher , 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b) While nine of these cantons already implemented the algorithm, more cantons are currently discussing the implementation too. In Thurgau, the initiative "Mehr Transparenz bei Wahlen im Thurgau" is currently pending at the Grand Council of the Canton and a special parliamentary committee is examining the iniatitive, also in the canton of Lucerne, the iniatitive "Faire Wahlen ohne Listenflut" was submitted and the voters will be able to decide on the effectuation (Grosser Rat des Kantons Thurgau, n.d.; Regierungsrat des Kantons Luzern, 2026, pp. 19–20). In the canton of St.Gallen in 2026 an initiative to adapt the Pukelsheimer algorithm was launched (Initiativkomitee «Damit jede Stimme zählt», n.d.).

The amount of cantons who have already implemented the system or are debating about doing so, suggests that the decision by the FSC had a substantial impact on how Swiss citizens elect their regional parliaments today. It seems that the double-Pukelsheimer through its wide implementation

across cantons is a success story. Since the first election under double Pukelsheim in Zurich in 2007, the system has been used in 32 cantonal parliamentary elections.[1]

Taken together, this number of elections and almost 20 years of experience provide a reason to evaluate the impact of the reform, especially on smaller parties. How much has the reform changed smaller parties' seat shares and how large is the actual difference in seat allocation between the Hagenbach-Bischoff and Double Pukelsheim system? Are there other major differences, for example based on the pre-reform district sizes and how do smaller parties seat shares evolve across successive elections?

Existing research primarily examines Double Pukelsheim in legal, normative, or system-level terms. A few studies also conduct counterfactual comparisons, but these cover either individual elections or only one election per canton (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592; Leuzinger, 2018, p. 42; Lüscher, 2024, p. 13). Although their results indicate that smaller parties can benefit from Double Pukelsheim, the magnitude and consistency of this mechanical allocation difference across successive elections remain less systematically examined.

This thesis addresses this gap by applying a consistent Hagenbach–Bischoff counterfactual to 31 elections in eight transition cantons. Two Graubünden elections are presented as contextual comparisons but excluded from the main transition estimate. Holding the observed post-reform votes and list structure constant, the analysis measures the difference in the share of seats won by smaller electoral lists and examines variation across institutional settings and successive elections.

4 Theory and Literature Review

4.1 Institutional and Theoretical Background

This chapter provides the institutional and theoretical background for analyzing the transition from Hagenbach-Bischoff to Double Pukelsheim and reviews the relevant literature. It first outlines the historical development of electoral systems in Switzerland and then explains the functioning and limitations of the Hagenbach-Bischoff and Double Pukelsheim methods. It subsequently examines the adoption and canton-specific implementation of Double Pukelsheim. Finally, the literature review situates this thesis within previous research before the expected effects of the transition are derived.

4.1.1 Historical Development

Until the late nineteenth century swiss voters elected their regional parliaments only under majority voting. In a majority voting system, the candidates or parties who receive the relative or absolute majority receive the mandates, which leads to a systematic favourment of the dominant parties. Over the time, this inequality, led to dissatisfaction among smaller parties. Subsequently, these demandend, to change to a proportional electoral system. In 1891 the canton of Ticino, was the first one to use such a PR-System to vote for their Grand Council, during the following 30 years multiple cantonsfollowed. When the socialist and conservative minorities on their third attempt in 1918 achieved, that the

[1]The 32 elections comprise Zurich (2007, 2011, 2015, 2019, and 2023; Amt für Statistik und Daten Kanton Zürich, 2026); Schaffhausen (2008, 2012, 2016, 2020, and 2024; Kanton Schaffhausen, n.d.; Staatskanzlei Kanton Schaffhausen, 2020, 2024); Aargau (2009, 2012, 2016, 2020, and 2024; Department für Finanzen und Ressourcen des Kantons Aargau, 2009; Regierungsrat des Kantons Aargau, 2013; Statistisches Amt Kanton Aargau, n.d.); Nidwalden (2014, 2018, 2022, and 2026; Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026); Zug (2014, 2018, and 2022; Kanton Zug, 2022a, 2022b, 2022c, 2026); Schwyz (2016, 2020, and 2024; Staatskanzlei Kanton Schwyz, 2016, 2020, 2024); Valais (2017, 2021, and 2025; Kanton Wallis, 2016, 2020, 2024; Staat Wallis, 2017); Uri (2020 and 2024; Standeskanzlei Uri, 2024a, 2024b); and Graubünden (2022 and 2026; Kanton Graubünden, 2022, 2026).

National Council will be elected through a proportional system, it boosted the general adoption of the system across Switzerland. Most of the cantons Grand Councils aswell as the National Council were then elected with the Hagenbach-Bischoff or another proportional system. In the beginning of the 21th century, the cantons of Graubünden and Appenzell Innerrhoden were the only ones to still vote their parliaments with the majority system. (Gilg, 2014, "Mehrheits- und Verhältniswahl" section ; Bundeskanzlei, 2013, p. 2) Nowadays the Kanton Appenzell Innerrhoden is the only remaining canton voting the Grand Council by majority voting (Bundeskanzlei, 2025, p. 18). It can therefore be said, that proportional systems have become the dominating electoral systems for cantonal parliaments in Switzerland.

In the last 20 years the prior dominance of the Hagenbach-Bischoff system has been demolished by the wide implementation of the Pukelsheimer Algorithm. As of 2025, eleven cantons distribute the parliamentary seats with Hagenbach-Bischoff and ten cantons with the DP, two cantons namely Glarus and Basel-Landschaft use the Sainte-Laguë method and two namely Ticino and Vaud use the Hare/ Niemeyer method (Bundeskanzlei, 2025, p.18). Both of the later named methods, are generally associated with higher proportionality than the HB system, therefore this might partly epxlain why none of the cantons using these two PR-Systems have adopted the Double Pukelsheim system (Bundeskanzlei, 2025, pp. 22; Deutscher Bundestag, 2007, p.9).This means that of the nine cantons analyzed here, eight previously used Hagenbach-Bischoof for at least the majority of their parliamentary seats and only Graubünden had used a different system with majority voting. With respect to, that Zug's Neuheim constituency and Uri's smaller municipalities also used Majorz rather than Hagenbach-Bischoff (Bundeskanzlei, 2013, Table 1; BGE 136 I 376 E. 5.1 ; Standeskanzlei Uri, 2020, p. 1; Statistisches Amt des Kantons Zürich, 2011, p. 286; Regierungsrat des Kantons Schaffhausen, 2007, p. 6; Regierungsrat des Kantons Aargau, 2007, p. 18).

4.1.2 The Hagenbach-Bischoff Method

To be able to comprehend later the differences between the HB and the DP Method one has to understand the allocation under the priorly used Hagenbach-Bischoff system.

Under Hagenbach-Bischoff, seats are allocated separately in each constituency. Let M denote the number of mandates in the constituency, v_i the votes for an eligible list i, and V the total list votes used to calculate the distribution number. Whereas the distribution number D is the smallest integer strictly greater than $V/(M+1)$. In the first allocation, each list receives the integer part of its votes divided by D (Bundeskanzlei, 2025, p. 41):

$$D = \left\lfloor \frac{V}{M+1} \right\rfloor + 1, \quad s_i = \left\lfloor \frac{v_i}{D} \right\rfloor.$$

If the first allocation leaves seats unassigned, the remaining seats are awarded one at a time. For each list, its votes are divided by the number of seats it currently holds plus one. The list with the highest quotient receives the next seat; its quotient h_i and the allocated seats s_i are then recalculated, and the process continues until all M seats have been filled:

$$h_i = \frac{v_i}{s_i + 1}.$$

It should be noted, that the HB system calculation produces the same allocation as the D'Hondt divisor method with downward rounding (Bundeskanzlei, 2025, p. 41). Further there are some factors, like list alliances, eligibility thresholds, and rules for resolving equal quotients which depend on the electoral law of the corresponding canton. The counterfactual in this thesis applies the relevant former cantonal rules within each constituency, as detailed in the Data Cleaning section.

The central weakness of the traditional cantonal application of Hagenbach–Bischoff does not arise solely from the allocation formula, but from applying it separately in constituencies that often have few seats. Small district magnitudes create high and across districts varying natural thresholds, meaning that votes for smaller parties may fail to contribute to a seat.This might hinder equality of electoral success and systematically disadvantage smaller parties. As shown in the later section on the adoption of Double Pukelsheim, these inequalities were central to the legal and political debates that led several cantons to adopt Double Pukelsheim. (Pukelsheim & Schuhmacher, 2004, pp. 507-508, 521; BGE 129 I 185 E. 7.1.1 and E. 7.6.2)

4.1.3 The Double Pukelsheimer Method

Based on the weaknesses of the Hagenbach-Bischoff Method and the FSC decision, the canton of Zurich comissioned the mathematician Friedrich Pukelsheim to develop a new method (Lüscher, 2024, p.10). The following section explores how the algorithm works and how it makes sure that it complies with the Federal Law and mitigates the HB weaknesses.

Based on Article 8 and 34 of the Federal Constitution, there are three dimensions of electoral equality, which have to be followed. These are principle of equal voting value, equality of voting power and equality of electoral value. The equal voting value ensures that each voter within an electoral district gets allocated the identical number of votes. The second principle is the equality of voting power (voting weight guarantee). This requires the population represented per seat to be approximately equal across electoral districts. Thirdly there is the equality of electoral success, which should assure, that all votes contribute equally to the election and that as many votes as possible are taken into account in the allocation of seats. (Pukelsheim & Schuhmacher, 2004, pp. 506-507) Districts with fewer than nine seats have a natural quorum above 10% under the approximation $1/(M+1)$, where M is the number of seats. The Federal Supreme Court treats 10% as a target for natural quorums when redesigning an electoral system. Exceptionally, compelling historical or regional minority-protection reasons may justify a higher natural quorum. (Pukelsheim & Schuhmacher, 2004, p. 505; BGE 131 I 74 E. 5.2–5.4) Equality of counting value concerns the formal treatment of votes, while equality of voting power depends on the separately determined ratio of population to seats in each electoral district (Pukelsheim & Schuhmacher, 2004, pp. 507, 515–516). For the traditional application of Hagenbach–Bischoff, the main difficulty lies in equality of electoral success when small districts impose high natural thresholds (Pukelsheim & Schuhmacher, 2004, p. 508; BGE 129 I 185 E. 7.1.1 and E. 7.6.2).

To approach this main issue of the Hagenbach-Bischoff, the Double Pukelsheimer or originally called the "Neues Zürcher Zuteilungsverfahren" knows a upper and a lower apportionment (Pukelsheim & Schuhmacher, 2004, p.513).

- **Upper Apportionment** To bypass the natural quorums, in the upper apportionment, the whole canton is seen as one single district. All the seats are then directly proportionally to the vote share allocated towards the parties. For that the votes from the district-level have to be summarized on a

cantonal level. This leads to the first difficulty, which arises through the variation in district magnitude and the resulting circumstance that voters receive across constituencies a different amount of votes. Therefore to be able to summarize the votes on a global level, the number of votes have to be translated, which is done as described followingly. The number of party votes per constituency are taken and divided with the respective number of seats. The resulting quotient gets rounded to the next number and indicates the amount of voters per list. The rounded quotients, can then be summed across all districts and divided by the cantonal divisor to get each parties total seats. The divisor itself is set by an official authority, in Zurich for example, its the Department of the Government Councillor which is responsible for the elections. The responsible authority, needs to assure that with the chosen divisor exactly as many seats can be allocated as there are available. The number of seats which result per party, are rounded again to the next number. (Pukelsheim & Schuchmacher, 2011, p.1584-1585 ; Kanton Zürich, 2003, § 10 and 103)

- **Lower Apportionment** This part of the apportionment process, assigns each party's cantonwide seats from the upper apportionment to its district lists while preserving each district's predetermined seat total. First, the party votes in each district are divided by a district divisor chosen so that the rounded results add up to the district's seats (Pukelsheim & Schuhmacher, 2011, p. 1585). These initial allocations may not match the parties' cantonwide entitlements: in the 2011 Zurich election, the SVP would have received 55 seats instead of the 54 assigned in the upper apportionment . To correct such differences, every district list of a party is divided by the same additional list group divisor. A divisor above one reduces the party's unrounded quotients and can remove seats while a divisor below one can add seats. (Pukelsheim & Schuhmacher, 2011, p. 1586) This effect can be illustrated with a hypothetical numerical example based on the 2011 seat mismatch for the SVP mentioned above. Now it could be that the SVP list in District Y would have a quotient after division by the district divisor of 2.51. With the standard rounding, the SVP would receive three seats in this district. But if now as a list group divisor 1.01 is taken, then this quotient falls to $2.51/1.01 \approx 2.49$, and the SVP would get allocated only two seats in this district. Through that the mismatch would be resolved, considering that in no other district, the allocated number of seats for the SVP would drop too. If this is the case, then instead of 1.01 a smaller divisor would had to be choosen until, only the amount of seats fall away, which are needed. Further might the change in District Y cause, that the district gets allocated one seat less then it should have. The party and district divisors are then adjusted in turn until every party receives its cantonwide entitlement and every district retains its fixed number of seats. (Pukelsheim & Schuhmacher, 2011, p. 1586)

To summarize, the upper apportionment in the Double Pukelsheim system reduces inequalities in electoral success caused by the high natural thresholds of small constituencies under the previous district-by-district Hagenbach-Bischoff system. It does so by pooling parties votes across the canton before seats are assigned to constituencies. This is also the main improvement of DP regarding the three dimensions of electoral equality in contrast to Hagenbach-Bischoff.

4.1.3.1 Trade-offs and Criticisms

Alike every proportional system, also the Double Pukelsheimer system has its flaws for which it gets criticized. One of the main weaknesses of the biproprtional apportionment compared to the HB System is its complexity. It gets mentioned throughout, that its relatively difficult to understand the system holistically and that there is profound knowledge needed, to be able to compute the seat allocation.

(Bundeskanzlei, 2013, p. 19 ; Bundeskanzlei, 2025, p.24 ; Lüscher, 2024, p.20) Lüscher tough mentions, that the usage of list alliances under the priorly used Hagenbach-Bischoff method also led to a high complexity situation. This especially since, beforehand the election it was impossible to estimate, which party would profit from the alliance. (Lüscher, 2024, p.20) Another main point of criticism, are the so called "discordant seat allocations". The discordant seat allocations describe the phenomena when within a district a party gets allocated more seats than another party with more votes in that district. This phenomena can appear through the lower apportionment process and the dual role of the system to achieve proportionality both on cantonal and district-level. The emerging situation, can cause confusement and frustration especially within the party which received more votes in that specific constituency. (Pukelsheim & Schuhmacher, 2004, p. 519) Not only for the parties themself it can be irritating, but also for the voters. This especially since in Switzerland elections often contain a personal vote component. If then a popular figure from one party, loses its seat to a less known figure from another party and district, this may raise concerns and lower the acceptance of thee system. A reaction to this flaw of the DP-System, was that the cantons of Zug, Schaffhausen and Graubünden implemented a majority clause, which aims to ensure that the strongest list in the corresponding constituency receives atleast one seat. (Lüscher, 2024, p. 21) Friedrich Pukelsheim was already aware of this issue in 2004 and pointed out that the so called majority requirement can in rare ocassions be not fulfilled. On base of this he mentioned the possibility of the implementation of such a majority clause. (Pukelsheim & Schuhmacher, 2004, p. 520) Another issue of the system, is that different turnout level and different proportions of residents who are eligible to vote in the consituencies can determine the allocated mandates of other constituencies. To illustrate this, Lüscher provides an example: There are three Districts with the same size and number of mandates, but while District 1 produces 50% of all votes, District 2 and 3 only produce each 25%. Through this District 1 determines 50% of the canton-wide party result, while only having 33.3% of the mandates. Subsequently this leads to under- and overrepresentation in the corresponding districts. (Lüscher, 2023, pp. 578-579) One other point which was often discussed to be a possible weakness of the Double Pukelsheim, is a possible Party System Fragmentation. For example both reports of the Swiss Federal Chancellery in 2013 and 2025 mention the possible party fragmentation in parliaments. As a possible measure to prevent fragmentation, the implentation of thresholds is mentioned. (Bundeskanzlei, 2013, p.18 ; Bundeskanzlei, 2025, p. 24) Despite this concerns, Sandro Lüscher came in his Thesis to the perception, that the general trend of a rising amount of parties in swiss cantonal parliaments, is unrelated to the implementation of the Double Pukelsheim. He then also mentioned, that the inexistency of an effect, might come from that several cantons have implemented the by the Federal Chancellerey mentioned thresholds. (Lüscher, 2024, p. 135) Therefore it can't be conclusively stated, whether the reform would have had an impact on party system fragmentation without the implentation of thresholds or not.

Taken together, these criticisms illustrate the central trade-offs of Double Pukelsheim. Although the system improves canton-wide proportionality and reduces the number of ineffective votes, it does so at the cost of greater computational complexity and potentially counterintuitive outcomes at the constituency level. Electoral thresholds can mitigate discordant seat allocations and party-system fragmentation, but may simultaneously weaken proportionality. The system therefore does not eliminate electoral distortions entirely, it rather prioritizes party proportionality and equal vote success over simplicity and fully autonomous constituency-level representation.

4.1.4 The Adoption of Double Pukelsheim in Swiss Cantons

In the following paragraphs the arguments brought up in each reform canton will be discussed in chronological order. The year in the beginning of each entry, marks the year of first application of the double Pukelsheimer in the corresponding canton.

- **2007:** The canton of Zurich as the initiatior of the development of the Double Pukelsheimer Algorithm, became the first canton to make use of it in 2007. The reform followed a 2002 voting-rights complaint about elections to Zurich's city parliament. The Swiss Federal Supreme Court partially upheld the complaint. It found that the thresholds in some districts with up to 33.3% were massively above the 10% threshold, which is seen as compatible with the prinicples of proportional representation. Further was it claimed, that thresholds vary to much between districts. The ruling required the city of Zurich to reform its electoral system, subsequently the canton of Zurich, which was facing the same issues as the city, implemented the Pukelsheimer Proporz for cantonal elections with a 5% district-level threshold for parties to enter parliament (BGE 129 I 185 E. 7.1.2, E. 7.6.2–7.6.3 and E. 8.3.2; Lüscher, 2024, pp. 9–10, 33-36 ; Pukelsheim & Schuhmacher, 2004, p.505). In 2021, Zurich added a 3% cantonwide threshold, first applied in the 2023 election. A change to how that threshold is calculated was adopted in 2025 and will first apply in the 2027 election (Regierungsrat des Kantons Zürich, 2024, pp. 2, 5; Regierungsrat des Kantons Zürich, 2025, p. 1).
- **2008:** Schaffhausen: In a 2004 popular vote, Schaffhausen's electorate approved a reduction of the cantonal parliament from 80 to 60 seats. Shortly after a motion calling to reform the electoral law to better protect smaller parties was submitted. After this the Schaffhausen Government proposed a reform of the electoral system and the implementation of the biproportional apportionment. The biproportional apportionment was also preferred, because of the Supreme Courts decisions in Zurich and Aargau. (Regierungsrat des Kantons Schaffhausen , 2007, pp. 1, 3, 13-14) Subsequently the parliament (2007) and the voters (2008) approved the changes. In 2015, the cantonal parliament added a majority clause, which aims to ensure that the strongest party receives at least one seat in the corresponding district. (Lüscher , 2024 , pp. 42- 43) In addition to the majority clause, did the canton of Schaffhausen decide to prohibit list alliances (Kanton Schaffhausen, 1904/2019, Art. 2b para. 3).
- **2009:** Aargau's voters in the year 2003 approved a proposal in a popular vote, which wanted to lower the amount of parliamentary seats from 200 to 140. This decision naturally raised the thresholds in the districts and therefore a second constitutional amendment allowing to merge small districts was proposed and approved by the voters. However, the parliament decided against making use of it. As a consequence of this decision, the electoral equity wasn't given anymore. The Government of Aargau had to act and proposed to the Grand Council to either merge districts or to change to the Double-Pukelsheim system. The Grand Council though, again decided not to change anything. Shortly after this parliamentary decision the FSC issued in a verdict, that there are existing inequalities, arising from the varying thresholds and district sizes. The court ordered that this needs to be changed. In September 2007, the Grand Council with a narrow majority then decided to implement the Double-Pukelsheimer. In 2011 a 5% local (district) and a 3% global quorum was added after approval of the Grand Council and the voters (Lüscher, 2024, pp. 38-40)
- **2014:** Nidwalden: In the year 2010, members of the Green Party in Nidwalden filed a complaint regarding the seat allocation process for the cantonal elections (Lüscher, 2024, p. 44). That complaint got rejected initially by the Cantonal Court but later the cantonal decision got overruled by the FSC.

The Supreme Court rejected the argument that historical practice justified the allocation method (BGE 136 I 352 E. 4.; Lüscher, 2024, pp. 44–45). As a reaction to this decision the parliament of Nidwalden came up with multiple possible reforms. The most popular of the suggested reforms, was the change towards the biproportional system. The parliament and the electorate subsequently approved the implementation of the Double-Pukelsheimer. (Lüscher, 2024, p.45)

- **2014:** Zug: In 2010, representatives of several parties formed a complaint at the FSC regarding the differently sized districts. The Federal Supreme Court agreed on the appeals main point (BGE 136 I 376 E. 4.5). As a reaction to this decision the cantonal Government presented three different possibilities on how to adapt to the FSC ruling. Of which the responsible comission favoured the adoption of the Double Pukelsheim. Subsequently, the parliament agreed on the adaption of the Double Pukelsheim and added based on the model of Aargau a 5% district-level and 3% cantonal-level threshold. (Lüscher, 2024, p. 47-51) A competing proposal to retain the existing system was struck down by the FSC in 2013 (BGE 139 I 195 E. 4). In addition to the thresholds, the canton also implented a majority clause, aiming to ensure that the most popular list within a district always receives at least one seat (Kanton Zug, 2006/2026, § 52e paras. 1a–1b). Voters approved the biproportional reform in September 2013 (Lüscher, 2024, p.51).
- **2016:** Schwyz: The reform process began with the approval of the new cantonal constitution by the electorate. In that it was foreseen, to keep the current electoral system, which means that each municipality is its own electoral district. (Lüscher, 2024, p. 53) In September 2011, the Cantonal Government prepared the Cantonal Council Election for 2012, against the decisions published there three individual complaints were raised at the FSC. The main complaint was that the thresholds in some constituencies were too high. The Court agreed on the complaints of the three parties partly and stated that the current electoral system doesn't meet the requirements of a proportional representation system (Urteil des Bundesgerichts 1C_407/2011 etc. vom 19. März 2012, Sachverhalt A–B und E. 5.3, 6). Later, the Federal Assembly, which is responsible for guaranteeing cantonal constitutions, also refused to grant its guarantee to the contested provisions of Schwyz's new constitution (Schweizerische Eidgenossenschaft, 1999, Art. 172 ; Lüscher, 2024, p.53). As a reaction towards this decision, two independent initiatives to resolve the issues were formed by political parties in the canton. One of them wanted the implementation of a majority system and the other argued for fairer proportional representation with the willingness to remove guaranteed seats for municipialities. The government reacted with the proposal to inaugurate the DP-System while keeping municipialites as districts and preserving their guaranteed seats. The parliament supported this counterproposal and decided to implement a 1% cantonal-level threshold. The electorate could then decide on either the majority vote initative or the counterproposal. The counterproposal and with that the implementation of the DP was approved. (Lüscher, 2024, pp. 53-55)
- **2017:** Valais: In the year 2004, a complaint at the FSC was filed regarding the varying thresholds in the electoral districts. By that time, the FSC rejected the complaint and argued that there are historical reasons for the shapes of the electoral districts. (BGE 131 I 85 E. 2.3–2.5) Six years later an alliance from the Upper Valais proposed a popular initiative named "Every Vote Counts" and suggested with that the implementation of the DP with a total of only three districts. In 2011 the canton then set up a commission to develop proposals in these areas. (Lüscher, 2024, pp. 58–59) In 2012 multiple voters again filed a complaint at the Supreme Court. The FSC then changed its standpoint and ruled

that the existing system violated the Federal Constitution (BGE 140 I 107 E. 4–5). Subsequently, the parliament approved a constitutional reform introducing biproportional seat allocation, with the addition of a 35 seats guarantee for the german-speaking Upper Valais. Voters narrowly rejected the package aswell as the initiative "Every Vote Counts" in 2015. To ensure lawful elections in 2017, the parliament adopted a temporary decree in March 2016 without the seat guarantee. The resulting system retained 14 subdistricts, grouped them into six larger districts, and applied biproportional allocation separately within each larger district. A year later, the Parliament incorporated these rules into the cantonal electoral law and with that made these arrangement permanent. (Lüscher, 2024, pp. 61–63) The Canton Valais further decided to implement a 8 % threshold which a party has to reach in atleast one of the 14 subdistricts to be included in the upper-apportionment allocation across six upper-level districts (Kanton Wallis, 2004b, Art. 154 Abs. 1).

- **2020:** Uri used a mixed electoral system: municipalities with one or two seats elected their representatives by Majorz, whereas those with at least three seats used proportional representation. A complaint against the seat allocation for the 2016 election led the Federal Supreme Court to rule that the allocation method within the Proporz municipalities violated electoral equality. The Court did not reject the mixed system itself, which it considered justifiable because it preserved personality-oriented elections and an autonomously elected representative for very small municipalities (BGE 143 I 92 Sachverhalt, E. 5–5.4 and E. 6.5 ; Kanton Uri, 2019, pp. 6–7). The cantonal government subsequently proposed Double Pukelsheim for all eight municipalities which had priorly used Hagenbach-Bischoff. During the legislative debate, however, the Landrat raised the threshold for the usage of the proportional system from three to five seats, moving Attinghausen, Flüelen, Seedorf, and Silenen to Majorz and leaving only Altdorf, Bürglen, Erstfeld, and Schattdorf in the Double Pukelsheim allocation. The parliamentary majority reasoned that a majority system is better suited for the four municipalities' small populations and strong autonomy. It also pointed to the weak role or absence of local party organizations, occasional uncontested elections, and electoral statistics suggesting that voters chose personalities rather than party lists. The government opposed the extension and warned of a high risk of failure, if it would come to another challenge of the electoral system in front of the FSC. Uri's voters though, approved both the constitutional amendment extending Majorz and the separate Proporzgesetz reform in May 2019 (Kanton Uri, 2019, pp. 4, 10–11; Lüscher, 2024, pp. 64–67). As of October 2026, Uri's revised electoral system remains in force and hasn't been overturned by the FSC.
- **2022:** Graubünden: Before the Implementation of the biproportional apportionment the canton was one of the last to use a majority based electoral system. This even though, that throughout the years there were multiple attempts to introduce a PR system. A main reason for these attempts to fail, was the fear that the protection of local minorities and the 20 single-seat-constituencies would be weakened. (Lüscher, 2024, p.69) In 2017 then multiple parties and individuals filed a complaint at the Federal Supreme Court after the cantonal government had released the seat allocations for the districts in the upcoming election. The court partially upheld the complaint and it especially stated that the seat allocation for the smallest district (Avers) and the six largest districts needed to be revised. Specifically the Court found that the guaranteed seats for Aver's 160 eligible voters violated equality of voting weight, while majoirtiy voting in the six largest constituencies unjustifiably impaired the equal effect of votes on seat allocation. (BGE 145 I 259 E. 7.3 and E. 8.5.3) As a response to

the FSC decision, the Government Council launched a consultation process after which, it presented nine different models. While the government preferred the DP-System, the municipalities and large parties favoured a mixed electoral model. The willingness of the Socialdemocrats to withdraw a separate initative helped build support for Double Pukelsheim. The parliament then chose the Double Pukelsheim as a compromise, since it made the seat allocation more proportional while keeping representation for small districts. A 3% canton-level threshold was additionally introduced. Moreover added the canton a majority clause similar to the one in Schaffhausen and Zug (Kanton Graubünden, 2021/2026, Art. 28 paras. 2–4). In 2021 the electorate voted in favor of implementing the DP-System (Lüscher, 2024, pp. 71-72).

- **2027:** Basel-Landschaft: Unlike other cantons, in Basel-Landschaft there were no implicit quorums of more than 10%, which means that the law complied with the FSC ruling. In 2015 though the parliamentarian Klaus Kirchmayr requested the cantonal government to examine the comparative effects of the current system with the Double Pukelsheimer. The Government didn't see any need for a reform, the parliamentarian though kept working on the topic and demanded the establishment of a special comission to review the topic. Although the parliament also rejected this proposal, it instructed its executive board to work with external experts to examine how the electoral system could better reflect party vote shares while preserving elected representatives' local ties. A few years later in 2022, the executive board came to the conclusion, that it recommended the implementation of the DP. Through the support of the parliament, the government was then instructed to draft a revision of the electoral law. (Lüscher, 2024, p.86) In the year 2024 the government then finally submitted the bill and the Cantonal Council voted in favour of it (Regierungsrat des Kantons Basel-Landschaft, 2024 ; Landrat des Kantons Basel-Landschaft, 2024, p. 1052). The reform was subject to a mandatory referendum and the canton's electorate voted in favour of the implementation in 2025 (Kanton Basel-Landschaft, 2025b). The canton further decided to implement a 3% canton-level and a 5% district-level threshold and stipulated that each constituency receives one guaranteed seat (Kanton Basel-Landschaft, 1981/2025, Art. 39a and Art.49 lit.a). The first election under the new system will be held in 2027 (Kanton Basel-Landschaft, 2026).

To summarize, in 8 of 10 reform cantons, a raised complaint at the FSC led to a later reform. Only in the cantons of Schaffhausen and Basel-Landschaft did the reform process take place without a Federal Supreme Court ruling. Nevertheless the canton of Schaffhausen's decision was directly influenced by the FSC's decisions in Aargau and Zurich. Throughout all complaints, the concern was electoral equality. In seven of them excessively high natural thresholds in proportional constituencies were central. While in Graubünden the concerns were mainly raised by the use of of the majority system for its large districts and the allocation of seats among constituencies. This demonstrates the influence of the 2002 Federal Supreme Court decision regarding Zurich's City Parliament.

Beyond their different paths to reform, the cantons also implemented Double Pukelsheim under substantially different institutional conditions. The number of seats included in the Double Pukelsheim allocation varies from 37 in Uri where the system applies only to four municipalities to 180 in Zurich (Standeskanzlei Uri, 2020, p. 1; Kanton Zürich, 2003, § 88). Moreover, some cantons introduced explicit local or canton-wide electoral quorums, whereas others did not, potentially limiting smaller parties' ability to benefit from the more proportional allocation method. These arrangements have also changed over time—for example through the later introduction of thresholds or majority clauses while

party competition and vote distributions vary between elections. This institutional variation motivates the exploratory comparisons below: whether smaller parties seat-share gains differ by the presence of an explicit quorum, and across successive elections within the same canton.

4.2 Literature Review

Building on the institutional mechanisms described above, the existing empirical literature generally finds that Double Pukelsheim improves vote–seat proportionality and benefits parties that were disadvantaged by small electoral districts. However, the studies differ substantially in their research designs, geographical coverage, and treatment of counterfactual electoral rules.

Pukelsheim and Schuhmacher provide an early counterfactual comparison based on the 2011 Zurich election. They apply the former Hagenbach–Bischoff method to the observed votes while assuming four hypothetical list alliances based on political orientations and historical alliances. Under this counterfactual, the larger SVP, SP, and FDP receive additional seats, primarily at the expense of smaller parties such as the BDP and Alternative List. Their calculation also indicates considerably greater inequality in the electoral success of votes under Hagenbach–Bischoff. The analysis nevertheless concerns only one election, and its results depend partly on assumptions about which list alliances parties would have formed. (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592)

Lüscher conducts a similar comparison for the 2016 Schwyz election. Under his Hagenbach–Bischoff counterfactual, which does not assume any list alliances, the dominant SVP and CVP would have benefited. Whereas the left alliance and especially the GLP benefited from the implementation of the Double Pukelsheim. Most notably, the GLP's three observed seats would have been reduced to zero under Hagenbach–Bischoff. (Lüscher, 2024, p. 13) This example illustrates the potentially substantial effect of the allocation rule on an individual smaller party, but it does not establish whether effects of this magnitude occur systematically across cantons and elections.

Leuzinger broadens the comparison by examining the most recent election in each of the seven cantons that had introduced Double Pukelsheim by 2018. His study is the closest precedent for the present analysis because it calculates hypothetical seat distributions under the former cantonal electoral systems. However, it covers only one election per canton and assumes that the list alliances observed in the final pre-reform election would have persisted (Leuzinger, 2018, p. 42). The present thesis does not carry these historical alliances forward. Consequently, the two counterfactual specifications can produce different Hagenbach–Bischoff seat allocations. This illustrates the sensitivity of counterfactual results to assumptions about unobserved alliance formation. Leuzinger also mentions that he observed, that mainly smaller parties profited from the transition and that around 10% of the mandates mechanically change the party, through the transition (Leuzinger , 2018, p.42).

Taken together, the counterfactual calculations by Pukelsheim and Schuhmacher, Lüscher, and Leuzinger constitute the closest methodological precedents for the present thesis. They apply a former allocation rule to votes observed under Double Pukelsheim and compare the resulting counterfactual seat distribution with the actual allocation. Their comparisons therefore hold the observed post-reform vote distribution constant and isolate the difference produced by the allocation rules.

The interpretation of this difference can be clarified using Fiva and Folke's framework for separating the mechanical and psychological effects of electoral reform. Fiva and Folke examine a reform of Norwegian municipal elections, first applied in 2003, that replaced D'Hondt with modified Sainte-

Laguë. (Fiva & Folke, 2016 , p.266) Their principal estimate of the mechanical reform effect compares the actual pre-reform allocation with a counterfactual application of the new formula to the same pre-reform votes. They also apply the former and new formulas to the same post-reform votes to examine how behavioral responses may alter the magnitude of the mechanical effect. (Fiva & Folke, 2016, pp. 270–272) The Swiss studies described above, and the present thesis, follow this latter computational approach: the former formula is applied counterfactually to votes observed under the new system. This measures the mechanical allocation difference conditional on the observed post-reform vote distribution, but it does not estimate how parties and voters would have behaved had the former system remained in place or identify the reform's complete causal effect.

Rather than recalculating individual elections, Lüscher's principal empirical analysis estimates how cantonal political outcomes would have developed without the reforms. He finds that Double Pukelsheim substantially reduced vote–seat disproportionality, with stronger improvements in cantons that previously had smaller average district magnitudes (Lüscher, 2024, pp. 129–143). However, Lüscher finds no significant increase in the effective number of parliamentary parties or electoral candidacies. Voter turnout initially declined slightly, but this effect diminished over subsequent elections. These findings distinguish the reform's immediate mechanical consequences from its more limited behavioural effects on parties and voters.(Lüscher, 2024, pp. 143–145)

The existing literature therefore provides strong evidence that Double Pukelsheim improves proportionality, as well as several examples in which smaller parties gain seats. What remains less systematically examined is the magnitude and consistency of the conditional mechanical allocation difference across successive elections. Existing counterfactual studies either focus on individual elections or cover only one election per reform canton, while Lüscher's analysis examines aggregate disproportionality and broader political outcomes rather than the election-specific seat gains of smaller parties.

This thesis addresses that narrower gap by applying a consistent Hagenbach–Bischoff counterfactual to 31 elections in eight transition cantons. Two elections from Graubünden are additionally presented as contextual comparisons. Because Graubünden previously used majority voting rather than Hagenbach–Bischoff, these allocations are hypothetical and are excluded from the main transition estimate. The thesis holds observed post-reform votes constant, applies a common definition of smaller parties, and measures their change in seat share for every election. In contrast to Leuzinger (2018) and Pukelsheim & Schuchmacher (2011), it does not assume that hypothetical or historical pre-reform alliances would have persisted. The analysis therefore estimates the mechanical allocation difference conditional on the observed votes without introducing an additional behavioral assumption about counterfactual alliance formation.

4.3 Expected Effects of the Transition

Based on the theoretical considerations developed in the preceding sections, several effects can be expected from the transition of the Hagenbach–Bischoff method to the Double Pukelsheim system. Under a district-by-district Hagenbach–Bischoff allocation, small district magnitudes create high natural thresholds for winning a parliamentary seat. Votes for parties which fail to win a seat in a district, won't contribute elsewhere. This produces ineffective votes and therefore disproportionately disadvantages smaller parties. The biproportional apportionment instead through the upper apportionment pools votes across the whole relevant area, before it distributes each party's seats among the

districts. This should increase the equality of electoral success and reduce vote–seat disproportionality. Therefore mechanically, and conditional on the same observed votes, smaller parties that surpass the applicable legal thresholds are expected to receive a larger seat share under Double Pukelsheim, while the seat advantage previously enjoyed by larger parties should decline (Pukelsheim & Schuhmacher, 2004, pp. 506–508, 513-514 , 518).

The magnitude of this mechanical difference should depend particularly on district magnitude under the Hagenbach–Bischoff comparison. The fewer seats allocated in a district, the larger the vote share a party generally needs to obtain mandates. Double Pukelsheim should therefore produce larger gains for smaller parties where the former or counterfactual Hagenbach–Bischoff allocation consists of many low-magnitude districts than where seats are already allocated in relatively large districts. This expectation is consistent with Lüscher's finding that the reduction in disproportionality was stronger in cantons with smaller pre-reform average district magnitudes (Lüscher, 2024, pp. 132–133).

The effect is also expected to depend on the specific institutional design adopted by each canton. Reflecting Switzerland's federal structure, the reformed cantons differ in their use of canton-wide and district-level legal thresholds, majority clauses, and other supplementary adjustments. Explicit electoral thresholds can restrict gains among smaller parties by excluding lists that do not reach the required vote share and can therefore partly offset the greater proportionality produced by the upper apportionment (Bundeskanzlei, 2013, p. 18; Lüscher, 2024, pp. 107, 135). The majority clause on the other hand as no direct effect on the smaller party seat-share gain. Since the cantons ensure in their law, that the seat entitlements according to the upper apportionment hold. Zug's law for e.g. explicitly states that the majority clause gets deactivitated if it is essential to fulfill the upper apportionment seat entitlement. (Kanton Schaffhausen, 1904/2019, Art. 2d para. 1bis and 2(b); Kanton Zug, 2006/2026, § 52e para. 1a, 1b and 2(b); Kanton Graubünden, 2021/2026, Art. 28 para. 2 and 4(b))

Valais and Uri are particularly important special cases because Double Pukelsheim does not cover the canton through a single canton-wide upper apportionment. Valais conducts separate upper apportionments in six electoral regions, so votes aren't pooled across the entire canton. This is expected to limit the proportionality-enhancing effect relative to a fully canton-wide implementation, particularly for parties whose support is dispersed across regions (Lüscher, 2024, p. 57). In Uri, Double Pukelsheim applies only to the four biggest muncipialities, while the remaining ones use Majorz. This limits the territorial and numerical scope of the reform, although it does not necessarily reduce the mechanical effect among the four municipalities included in the upper apportionment (Kanton Uri, 2019, pp. 4, 10–11; Lüscher, 2024, pp. 64–67).

Comparisons across successive elections also provide exploratory context by examining whether smaller parties' mechanical seat-share gains persist or vary over time. No directional trend is hypothesized, as vote distributions, the composition of the smaller-party group, and electoral rules can change between elections

The estimated difference may additionally depend on the electoral rules used to construct the counterfactual outcomes, including the treatment of list alliances and canton-specific allocation provisions.

Together, these expectations provide the foundation for the following hypothesis and exploratory questions concerning differences in smaller parties' seat shares.

5 Hypotheses

To answer the research question, the thesis tests one main hypothesis concerning how the transition from the Hagenbach–Bischoff system to Double Pukelsheim mechanically affects the seat shares of smaller parties. Complementary exploratory analyses evaluate whether this effect varies with district magnitude, the presence of an explicit electoral threshold, and successive elections since implementation within each canton. Together, these analyses identify both the immediate mechanical and potentially evolving consequences of the electoral-system reform.

5.1 Hypothesis 1

The Hypothesis 1 researches whether smaller parties when holding votes constant collectively receive a larger seat share under Double Pukelsheim than under the conterfactual Hagenbach-Bischoff allocation.

For the analysis the following preqreuisites and decision are taken. In each election e, smaller parties or electoral lists are those with less than 10% of the vote in the *analyzed area*. The variable *analyzed area* describes the whole canton except in Uri, where the *analyzed area* includes only the four municipalities using the DP-System. The 10% cutoff is chosen with reference to the Swiss Federal Supreme Court's benchmark that natural (implicit) thresholds should not exceed 10% when an electoral system is redesigned (BGE 131 I 74 E. 5.3–5.4; Bundeskanzlei, 2013, p. 8). This benchmark motivates the definition of smaller parties used in this analysis. As a sensitivity check, the 10% threshold is complemented by an analysis with a 8% and 12% threshold.

Let $S^{DP}e$ and $S^{HB}e$ denote the aggregated number of seats allocated to smaller parties in election e under Double Pukelsheim and the counterfactual Hagenbach–Bischoff system, respectively. Aswell as let M_e denote the total number of seats included in the comparison. Smaller parties' combined gain in seat share, shown for each election in Figure 1, is calculated as follows:

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points} .$$

The hypothesis uses the unweighted mean, $\bar{\Delta} = \frac{1}{E}\sum_{e=1}^{E}\Delta_e$, so each election counts equally. The analysis also reports a mean weighted by the number of analyzed seats in each election for comparison.

This leads to the following directional hypothesis: $\boldsymbol{H_1}$**:** Across the analyzed elections, the mean mechanical difference $\bar{\Delta}$ is positive: electoral lists or list groups with less than 10% of the vote receive, on average, a larger share of the analyzed seats under Double Pukelsheim than under the counterfactual Hagenbach–Bischoff allocation.

As a supplementary and exploratory analyis, it's examined how individual parties seat-share gains vary across vote-share groups. Here no directional hypothsis regarding the ordering of the groups is formulated. Instead the comparison provides context for the aggregate resulty by showing which party-sizes experience higher and lower seat-share gains.

5.2 Exploratory Questions

5.2.1 Exploratory question 1: Differences by district magnitude

The first exploratory question researches on the differences by district magnitude. Concretely it asks: Does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual

Hagenbach–Bischoff allocation tend to be larger in cantons whose initial district structure was more fragmented? All district measures use a fixed historical baseline from each canton's last pre-reform election, with Schaffhausen adjusted to 60 seats. District size is measured with multiple parameters. The primary measure is the mean baseline district magnitude. Moreover is the median, the relative share of districts with less than 9 seats and the coefficient of variation between the districts reported. These additional measures add context and can reveal otherwise unobserved patterns for e.g. may the mean be influenced by a small number of unusually large or small districts. For canton c, with $D_{c,0}$ baseline districts and district magnitudes $m_{dc,0}$, the mean district magnitude is

$$\bar{m}_{c,0} = \frac{1}{D_{c,0}} \sum_{d=1}^{D_{c,0}} m_{dc,0} = \frac{M_{c,0}}{D_{c,0}}.$$

A lower baseline mean or median district magnitude is expected to be associated with a larger average gain for smaller parties. As a complementary measure, the analysis reports the baseline share of districts with fewer than nine seats,

$$q_{c,0} = \frac{1}{D_{c,0}} \sum_{d=1}^{D_{c,0}} \mathbb{1}(m_{dc,0} < 9).$$

The nine-seat cutoff follows the Swiss Federal Supreme Court's reasoning: under the natural-quorum approximation $1/(M+1)$, districts with fewer than nine seats impose a threshold above 10%. A larger baseline value of $q_{c,0}$ is therefore expected to be associated with a larger average gain (BGE 131 I 74 E. 5.3–5.4 ; Bundeskanzlei, 2013, p. 8). Because the number of transition cantons is small and their electoral rules differ, these relationships are interpreted descriptively. Graubünden is because of the priorly stated reasons only shown for context.

As a complementary measure, EQ1 examines inequality between district sizes using the coefficient of variation (CV). While the median captures the size of the typical district, CV describes how widely district magnitudes differ relative to their mean. The CV is calculated followingly:

$$CV_{c,0} = \frac{\sqrt{\frac{1}{D_{c,0}} \sum_{d=1}^{D_{c,0}} (m_{dc,0} - \bar{m}_{c,0})^2}}{\bar{m}_{c,0}}.$$

Here, c denotes the canton and 0 its pre-reform baseline; $D_{c,0}$ is the number of districts, $m_{dc,0}$ their seat magnitudes, and $\bar{m}_{c,0}$ their mean. The numerator is the population standard deviation, using $D_{c,0}$ as the denominator of the variance because the analysis covers the full set of districts in the analyzed area.

The motivation to include CV as a measure is based on the FSC reasoning in BGE 136 I 352 E. 3.5, where it identifies unequal district seat counts as problematic for the equality of electoral success. Equal to the other additional measures, this measure is shown to provide additional insight and not to test a specific hypothesis.

5.2.2 Exploratory question 2: Explicit electoral quorums

How does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation vary between elections conducted with and without an explicit electoral quorum?

An explicit quorum is a legally specified minimum vote share that a party or list group must reach at at a district or cantonal level to qualify for seat allocation. Among the eight cantons of the main analysis, five have applied an explicit quorum. Whereas Aargau is a special case, since it implemented the quroum only from the second election under Double Pukelsheim on. The canton of Graubünden implemented an explicit quroum and is shown for context. Further details around the details of the implementation of explicit thresholds across the cantons are stated in Section 4.1.4. Unlike the natural threshold created by limited district magnitude, the explicit quroum directly excludes parties that do not satisfy the legal requirements.

Because such quorums may prevent smaller parties from benefiting from the pooling of votes under Double Pukelsheim, the following directional expectation is formulated:

On average, the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation is smaller in elections with an explicit electoral quorum than in elections without one.

This comparison is interpreted descriptively because the level and geographical application of the quorums differ across cantons and elections and coincide with other institutional differences.

5.2.3 Exploratory question 3: Change across successive elections

How does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation evolve across successive elections within each canton?

This analysis extends Hypothesis 1 by following the seat shares under both allocation rules and their difference Δ_e from each canton's first election under Double Pukelsheim onward. It assesses whether the overall difference identified in Hypothesis 1 is persistent across elections or driven by particular election-specific outcomes.

Stable positive values of Δ_e would indicate a persistent mechanical advantage for smaller parties, whereas substantial fluctuations or changes in sign would suggest that the effect is strongly dependent on the circumstances of individual elections. The within-canton perspective further provides a clear separate image for each canton, without mixing their different electoral systems. Given that the smaller parties seat share, the party composition, aswell as the district magnitudes and electoral rules can change between elections, no directional trend can be hypothesized.

6 Research Design and Analysis

6.1 Research Design and Counterfactual Strategy

This thesis uses a deterministic counterfactual reallocation design. The main analysis covers all 31 cantonal parliamentary elections held under Double Pukelsheim in Switzerland through 2026 in the eight cantons in which Hagenbach–Bischoff previously was in use as the electoral system. The two elections from Graubünden are included as contextual comparisons since the canton had previously used a majority voting system. At the computational level, the observations are electoral lists within electoral constituencies and elections, while the primary unit of comparison is the election.

Each election is evaluated under two allocation regimes. The observed outcome is the official allocation under Double Pukelsheim, obtained from the relevant cantonal election authority. The counterfactual outcome is obtained by applying the specified former legal allocation rules to the same election's

observed votes, electoral units, seat magnitudes, and reported list structure. Each election therefore serves as its own comparison.

The counterfactual is constructed in five steps.

First, the official election files are imported and standardized to obtain, for every election, electoral unit, and reported list or list group, the observed votes, seat magnitude, and official Double Pukelsheim seats. Party and list labels are harmonized across source files, and voter numbers or vote shares are reconstructed where not already directly published by the structure of the official data.

Second, the counterfactual inputs are defined by holding the observed votes, electoral units, seat magnitudes, and reported list configuration constant. Joint lists reported as one unit in the official results remain pooled while lists affiliated with the same party, but running independently remain separated.

Third, the specified former allocation regime is applied separately within each electoral unit. Most counterfactual allocations use the district-level Hagenbach–Bischoff method described above. Where the former cantonal rules contained additional rules such as, quoruoms, majority-clauses, or tie-breaking provisions, these are incorporated into the calculation. The applicable canton-specific rules and features are documented in Section 6.2.

Fourth, the counterfactual seats obtained in the individual constituencies are aggregated to the party-election level and merged with the official Double Pukelsheim seat totals.

Finally, the resulting datasets are validated by checking party coverage, vote and seat consistency, electoral-unit seat totals, and the total number of seats allocated under both regimes. Where the source data provide multiple measures of the official Double Pukelsheim result such as upper-apportionment totals, lower apportionment totals, reported list mandates, or elected-candidate counts these measures are additionally reconciled. If exact ties between parties occur, they are if possible resolved with the applicable former cantonal laws. For the two cases in which the tie persisted, both possible seat allocations were presented for the affected parties. This assures, that a seat is never assigned unintentionally through e.g. the arbitrary ordering of rows. The two mentioned unresolved alternatives were observed in Graubünden 2022 and Schwyz 2024 and concern only parties above the 10% threshold and therefore do not affect the main smaller-party comparison.

The resulting contrast between the observed Double Pukelsheim allocation and the countrefactual is interpreted as a mechanical allocation difference conditional on the observed post-reform votes and not as the reforms complete causal effect. This since these votes were cast under Double Pukelsheim and the analysis cannot determine how voters, parties, number of candidates, campaign strategies, or list alliances would have differed if the former electoral regime had remained in force.

6.2 Counterfactual Assumptions and Scope

Applying the general counterfactual strategy requires several canton-specific decisions concerning electoral rules, list alliances, electoral units, and seat magnitudes. This subsection explains these decisions and defines the scope of the comparison.

Across the nine reform cantons, the counterfactual applies a district-level allocation to the electoral units retained from each observed election. In eight cantons, this generally reconstructs the formerly

applicable Hagenbach–Bischoff regime, subject to the exceptions summarized in Table 1.[2] In Graubünden, which previously used Majorz, the Hagenbach–Bischoff allocation is entirely hypothetical and is therefore excluded from the main transition estimate. In every case, the observed votes, electoral-unit seat magnitudes, and reported list structure are held constant. The common Hagenbach–Bischoff procedure is described in Section 4.1.2. This subsection therefore focuses on canton-specific adaptations and limitations of the reconstruction.

Importantly, the comparison in this thesis concerns the complete election-specific allocation regimes previously used in the cantons, rather than only the mathematical difference between the two apportionment algorithms. The observed outcome incorporates the Double-Pukelsheim rules applicable in the respective election, including any explicit electoral quorum. Further the counterfactual applies the relevant former district-level rules to the observed votes and electoral structure. Consequently, differences in seat allocation may reflect both the change in apportionment method and associated differences between the applicable rule packages.

[2] Aargau (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b); Staatskanzlei des Kantons Aargau, 2004, section "Neu: 140 statt 200 Mitglieder"); Graubünden (Kanton Graubünden, 2003/2018, Art. 27; hypothetical HB specification based on Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)); Nidwalden (BGE 136 I 352, E. 3.2); Schaffhausen (Regierungsrat des Kantons Schaffhausen, 1979, §§ 2, 36–37; Regierungsrat des Kantons Schaffhausen, 2007, pp. 2, 6); Schwyz (Urteil des Bundesgerichts 1C_407/2011 vom 19. März 2012, E. 5.2); Uri (Kanton Uri, 2018, Art. 22–23, "Bisher"; Kanton Uri, 2019, pp. 4, 11); Valais (BGE 140 I 107, Sachverhalt, pp. 108–109; Kanton Wallis, 2004a, Arts. 154–156); Zug (Kanton Zug, 2006, §§ 46–47, 55–56; BGE 136 I 376 E. 4.2, 4.4 and 5.1); and Zurich (Bentz, 2003, p. 2; Kanton Zürich, 1983/2005, §§ 87–88b)

Table 1: Former electoral systems and counterfactual treatment by canton.

Canton	Pre-reform allocation	Counterfactual treatment	Canton-specific counterfactual rule
AG	HB in 11 electoral districts	Former HB allocation reconstructed	Standard HB procedure
GR	Majorz	Hypothetical district-level HB allocation in 39 electoral districts ; excluded from the main estimate	Hypothetical standard HB specification based on the basis Forumula in Aargau Law
NW	HB in 11 municipalities	Former HB allocation reconstructed	Standard HB procedure
SH	HB in 5 multi-seat districts, majority rule in 1 single-seat district	Former mixed district-level system reconstructed	Ceiling-based distribution number; relative majority in the single-seat district; statutory tie-breaking where required
SZ	HB in 30 municipalities	Former HB allocation reconstructed	Standard HB procedure; unresolved exact ties retained as alternatives
UR	HB in 8 municipalities, Majorz elsewhere	HB reconstructed for the 4 municipalities using DP	Tie-breaking by initial remainder and then party votes; unresolved alternatives retained
VS	HB in 14 districts	Former district-level HB regime reconstructed	8% district eligibility threshold: ineligible lists' votes remain in the distribution-number calculation
ZG	HB in 10 municipalities; Majorz in Neuheim	HB applied in all 11 municipalities; hypothetical for Neuheim	Standard HB procedure; hypothetical HB application in Neuheim
ZH	HB in 18 electoral districts	Former HB allocation reconstructed	Tie-breaking by party votes, followed by lot; unresolved lots retained as alternatives

Of all canton-specific provisions in Table 1 several do not affect the smaller-party outcomes. For example, were the statutory tie-breaking rules in Schaffhausen, Uri, and Zurich not triggered in any analyzed election. The Schaffhausen rule for calculating the distribution number though, produced differences in two cases: 5,022 rather than 5,023 in Klettgau in 2008 and 2,196 rather than 2,197 in Neuhausen in 2024. These differences changed neither the initial nor the final seat allocation. In Schwyz 2024 and in Graubünden 2022, an unresolved exact tie produced alternative HB allocations for parties above the 10% threshold. Thus it only affected individual party totals but not the aggregated seat total of smaller parties used in the main analysis. The relative-majority allocation in Schaffhausen's

single-seat district was nevertheless applied and remains part of the reconstructed fomrmer system.[3]

Beyond the statutory allocation rules, the counterfactual requires an assumption about unobserved list alliances. Unlike Leuzinger and Pukelsheim & Schuhmacher, this analysis does not carry historical or hypothetical alliances forward. List alliances are in this thesis seen as strategic and election-specific rather than fixed institutional parameters. This leads to, that election-specific, historical alliances cannot simply be assumed to persist across subsequent elections with different parties and vote distributions. In Zurich, for example, the last Kantonsrat election under Hagenbach–Bischoff took place in 2003. Assuming that the alliances formed in 2003 would still have applied in 2023 despite two decades of changes in the party system would therefore introduce a strong and unverifiable behavioural assumption. Some parties that formed list alliances in Zurich in 2003 no longer existed or contested the 2023 election. Conversely, parties such as the Green Liberal Party, which contested the 2023 election, did not yet exist in 2003. (Statistisches Amt des Kantons Zürich 2003, table 3; Statistisches Amt des Kantons Zürich 2026; Grünliberale Partei Kanton Zürich n.d.). Therefore the counterfactual preserves joint lists or list groups represented as a single unit in the observed election data. But it does not reconstruct historical alliances or assume new alliances between separately reported units. This provides a transparent counterfactual that holds the observed electoral configuration constant and estimates the mechanical allocation difference between the specified election-specific regimes.

The exclusion of unobserved list alliances is an important limitation of the counterfactual. Under Hagenbach–Bischoff, list alliances can pool votes during seat allocation and thereby improve the chances of representation for smaller lists (Pukelsheim & Schuhmacher, 2011, p. 1590). If the parties observed under Double Pukelsheim would have formed such alliances under the counterfactual HB regime, excluding them may understate smaller parties' counterfactual HB seat totals and consequently overstate their estimated gain under Double Pukelsheim. However, neither the composition of these hypothetical alliances nor the distribution of any resulting seat gains can be reliably determined. Alliance gains could accrue to larger partners or leave the aggregated representation of parties below the 10% threshold unchanged. The direction is therefore theoretically plausible but cannot be quantified uniformly across elections.

A second research-design choice concerns the electoral units and their seat magnitudes. In most cantons, the reform retained the former territorial districts or municipalities, meaning that the observed electoral units broadly correspond to those previously used under HB. The principal scope deviations are addressed in the special-cases callout below. Regarding the district magnitudes, the use of election-specific magnitudes is deliberate. District seat entitlements are generally based on population and periodically recalculated. Consequently, the magnitudes from the last pre-reform election would not necessarily have remained applicable in later elections even if Hagenbach–Bischoff had been retained. The counterfactual therefore holds each observed election's electoral units and seat magnitudes constant across the two regimes. It asks how the same votes would have been translated into seats under the former district-level rules, conditional on the contemporary distribution of seats among districts.

The remaining deviations from this general approach are summarized below.

[3]These findings are based on the author's calculations of the official election data described in Appendix 1. Sources for the applicable legal provisions are provided in the preceding footnote.

ℹ Treatment of the Special Cases

Graubünden. The HB allocation is hypothetical and applied using the common allocation procedure employed elsewhere in the analysis. It's basic formula is taken from the former Aargau Law (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)). Thus the observations are shown in grey and excluded from the main transition estimate. In the 2022 election, appears an unresolved exact tie between the "FDP" and "SP_GRUENE" in Rheinwald, to resolve it an hypothetical lot would be needed. The tie, which doesn't affect the aggregated seat total of smaller parties, is therefore retained as an unresolved amlternative in the counterfactual.(Kanton Aargau, 2004, § 13; author's calculations using Kanton Graubünden, 2022) Robustness checks using Zurich's party-vote tiebreaker and Schaffhausen's ceiling-based quota preserved all possible allocations across the 39 districts in both 2022 and 2026 (Kanton Zürich, 1983/2005, § 88b; Regierungsrat des Kantons Schaffhausen, 1979, § 36).

Zug. Neuheim used Majorz before the reform. An exact Majorz counterfactual cannot be reconstructed because the results of a hypothetical second ballot are unobservable. Neuheim is therefore included using a hypothetical HB allocation.(BGE 136 I 376 E. 5.1; Kanton Zug, 2022a, 2022b, 2022c, 2026) The specific implementation approach adopted for Neuheim has no bearing on the main analysis.

Uri. Only Altdorf, Bürglen, Erstfeld, and Schattdorf used Double Pukelsheim in the elections analyzed here. They account for 37 of the Landrat's 64 seats (Standeskanzlei Uri, 2020, p. 1). Uri's vote shares, including the less-than-10% classification of smaller parties, are therefore based only on the four municipalities using Double Pukelsheim, not on the whole canton.

Valais. Unlike in the other cantons, upper apportionment is conducted separately within six electoral regions rather than canton-wide. Seats and votes are aggregated at the cantonal level only for this analysis; the actual allocation occurs independently within each region (Lüscher, 2024, p. 57).

Schaffhausen. The parliament was reduced from 80 to 60 seats when Double Pukelsheim was introduced. The counterfactual retains the 60-seat structure because comparing the two allocation methods requires the same number and distribution of seats (Regierungsrat des Kantons Schaffhausen, 2007, pp. 1-2, 6).

With the counterfactual rules and scope established, the next subsection describes the official election data and their standardization across cantons.

6.3 Data Sources and Processing

This subsection describes the official election data used in the analysis and explains how the heterogeneous cantonal source files were transformed into a harmonized dataset.

The official election results were obtained primarily from publicly available datasets published by cantonal state chancelleries and statistical offices. Where the data wasn't available online in structured formats, such as CSV or XLSX, they were requested from the responsible cantonal authorities directly. All contacted authorities except the Canton of Schaffhausen provided data in either CSV or XLSX. Therrefore the Schaffhausen results had to be extracted from official PDF documents. The precise

sources and elections covered are reported in Appendix Table 4. The granularity and organization of the source data vary considerably across cantons and elections, ranging from canton-wide files to separate district- or municipality-level results. In Schwyz, for example, the 2016 results were distributed across 30 municipality-specific excel workbooks, whereas the 2020 and 2024 results were each provided in a single workbook. Further did the structure of the files differ, some such as the Graubünden ones, were prepared for public or media use and through that readily accessible, while others had originally been designed for internal administrative calculations and where therefore less structured.

Party and list names were standardized using readable, clear and consistent abbreviations. The same political organization was assigned a common identifier across cantons and elections, some of the examples include "SVP", "GLP", "ALG", and "PIRATEN". Meaningful changes over time were nevertheless retained. In particular, "CVP" and "BDP" remain separately identified before their merger, while "DIEMITTE" is used only after the formation of the new party in 2021 (H.-P. Schaub, 2022). In bilingual Valais, French- and German-language labels referring to the same party were harmonized, generally using the German abbreviation for consistency with the other cantons. However, "SP_GAUCHE_CITOYENNE" and "SP_GRUENE_UNABHAENGIGE" were retained as distinct identifiers because they represented separately reported lists with partly different organizations in the background.

The treatment of joint and affiliated lists follows the electoral units reported in the official results. Where several political organizations contested an election on a single joint list, that list was treated as one analytical unit. Examples include "SP_GRUENE" in Graubünden in 2022 and "FGA_AL_PDA" in Zurich in 2023. It's important to state that joint lists and list alliances used under Hagenbach-Bischoff differ fundamentally. When forming a joint list, parties run together as a single electoral list (party), while under list alliances the parties run separately and the votes of the alliance parties are pooled later for the seat-allocation. (Pukelsheim & Schuhmacher, 2011, p. 1590) In addition to joint lists, in some cantons also affiliated or special-interest lists ran as a independent list. This phenomena occurred especially strong in Schaffhausen were among others the following affiliated lists ran for the 2016 election : "SVP_KMU" (SVP's small and medium enterprises list),"SVP_SEN" (SVP's senior's list), "JUSO" (Young Socialists) and "JFSH" (Young Liberals) (Kanton Schaffhausen, n.d.). Since these lists entered the official Double Pukelsheim allocation separately, they were also retained as separate units in the analysis. The analysis therefore contains exactly the lists which ran in the corresponding election with the effect that this decision possibly affects the measured number and seat share of smaller parties.

Beyond the harmonization of party and list names, the cleaning process involved restructuring vote data, identifying the relevant electoral units, and assigning their election-specific seat totals. The source files were converted into comparable tables containing party votes and the seat magnitude of each electoral district. Where parties appeared in separate columns, as in Aargau, Graubünden, and Nidwalden, these columns were reshaped into the variables *parties* and *party_votes*. Candidate-level exports required a different treatment. In all four Zug and Schwyz's 2020 election, each candidate row repeated the corresponding list's vote and seat totals. The scripts first checked that these repeated values were consistent and then retained one observation per municipality and list, preventing the totals from being counted multiple times. Source variables such as *list_votes* and *election_mandates*

were thereby converted into *party_votes* and *district_seats*. The geographic preparation depended on the structure of the official files. In Zurich, municipal party votes were summed within the electoral districts identified by the source variable *Geschaeft*. Aargau's files already contained district totals, which were selected using the *DISTRICT* classification. Similarly, the Graubünden script extracted the existing district totals by carrying each *Wahlkreis* heading down to its corresponding *Total* row. Municipalities were retained as allocation units in Nidwalden, Schwyz, and Zug. Their names were extracted from the source tables or, for Schwyz in 2016, from the municipality-specific filenames. Uri's script explicitly selected the workbook blocks for Altdorf, Bürglen, Erstfeld, and Schattdorf. In Valais, the 14 districts were assigned to six electoral regions in R according to the official electoral structure, preserving the different geographic levels required for the district-level HB counterfactual and the regional DP upper apportionment.

District seat magnitudes were obtained from the available official information for each election. In Aargau, district names were matched to officially sourced seat vectors or separate seat-allocation tables provided by the canton officials. Valais used explicitly recorded district magnitudes for each election year. In Zurich and Graubünden, magnitudes were reconstructed by counting elected candidates within each district. In Uri and Schwyz's 2016 election, they were calculated by summing the official DP seats within each municipality.

The shared finalization function then produced one observation per standardized party or list code and election. It created an election identifier combining the canton abbreviation and election year, such as *ZH_23*, and joined the HB and DP seat totals by party. The final columns were *election_id*, *parties*, *vote_share*, *small_party*, *seats_hb*, *seats_dp*, *parliament_size*, and *seats_allocated_dp*. The last two variables distinguish the full parliament from the seats covered by the comparison, this because of the Canton Uri, where these values are 64 and 37, respectively. District identifiers, raw party votes, and district magnitudes served as intermediate inputs rather than columns in the final combined dataset.

Vote shares were derived using the information available in each canton's official election files. In Aargau, Zurich, Zug, and Valais, party votes were divided by the corresponding district's seat magnitude to obtain voter equivalents. These quotients were rounded to the nearest integer, with half values rounded upwards, before being summed by party across districts. Each party's vote share was then calculated as its aggregated voter number divided by the total voter number of all included parties, multiplied by 100. Nidwalden instead used the voter numbers supplied in the official upper-apportionment worksheet. Schaffhausen used the published voter numbers extracted from the official result PDFs. Further where there the reconstructed voter numbers compared with the published values using election-specific tolerances to accommodate differences in reporting precision. Other datasets supplied vote shares directly or required different denominators. In Schwyz, the 2016 shares were calculated from each party's total cantonal votes relative to the sum of votes for all retained party lists, excluding the category for ballots without a party designation. For 2020, the denominator was the total accounted votes, counted once per municipality and checked against the summed list votes. The 2024 Schwyz shares and the Graubünden shares were taken directly from the official cantonal summaries. Uri's published shares refer exclusively to the four relevant municpialites and could therefore be retained. The shared finalization function rounded all vote shares to four decimal places.

The Double Pukelsheim results were assembled separately within the canton-specific scripts from official seat tables, municipal allocations, or elected-candidate records. Valais required a separate reconstruction of the upper apportionment within its six electoral regions, followed by comparisons with published official results. Each script supplied its election-specific DP results to the shared finalization function, which joined them to the HB counterfactuals by standardized party identifier. Validation routines checked electoral-unit coverage and seat totals and, where alternative representations were available, reconciled candidate counts or aggregated lower allocations with reported list mandates or upper-apportionment totals. District-level consistency checks and final allocation totals were applied according to the available source structure. Uri's allocation checks covered the 37 proportional seats included in the analysis, while the full parliamentary size remained recorded as 64. Unresolved HB ties were retained as alternative allocations.

One of the consistency checks revealed an error in the Zurich 2011 election data. There the party-votes total for the Socialdemocrats, were dobuled across all constituencies. The mistakes was found through the comparison of the summed party votes with the reported unit totals and cross-checking candidate and supplementary votes. Followingly, the error was reported to the Statistical Office of the Canton of Zurich. They confirmed the mistake and resolved it, the underlying cause was that a duplicate SP entry in the list-to-party mapping table had caused double counting during the aggregation of the official municipal results. The corrected dataset was uploaded and is used in this analysis. (S.Graf, personal communication, 2026-09-11)

All processing was implemented in the canton-specific R scripts in *data_cleaning_cantons*, supported by the shared functions in *00_functions.R*. Each *process_canton_*()* function combined its cleaned elections into a cantonal dataset. These nine datasets were subsequently appended row-wise using *bind_rows()* to create *all_elections*, preserving one observation per standardized party or list and election. Detailed procedures, source-specific exceptions, and validation arrangements are documented in Appendix 1, with the cleaning code provided in Appendix 2.

6.4 Measures and Analytical Strategy

The *analyzed_area* comprises the whole canton except in Uri, where it covers only Altdorf, Bürglen, Erstfeld, and Schattdorf, the four municipalities using Double Pukelsheim. These municipalities account for 37 of the canton's 64 parliamentary seats. The variable *seats_allocated_dp* records the number of seats covered by the comparison, while *parliament_size* records the full size of the cantonal parliament. These values are equal in all analyzed cantons except Uri.

For the purposes of this thesis, a smaller party is defined as an electoral list or list group with a vote share below 10% in the analyzed area. Youth lists and other affiliated lists are treated as separate analytical units whenever they enter the official Double Pukelsheim allocation separately. Accordingly, the term *party* refers to the standardized electoral list or list group recorded in the dataset and does not necessarily correspond to a broader organizational party family. *Smaller parties* are identified separately for each election using their observed vote shares. The same set of smaller parties is then used when comparing the two seat allocations within that election.

The main analysis compares the combined seat shares of smaller parties within each election. For this purpose, the seats of all lists below the 10% vote-share threshold are summed separately under Double Pukelsheim and Hagenbach–Bischoff. Each total is divided by the number of analyzed seats,

recorded in *seats_allocated_dp*, and multiplied by 100 to obtain the corresponding seat share as a percentage. This produces one aggregate seat-share difference per election. Using *seats_allocated_dp* as the denominator ensures that the measure refers to the same area as the votes and seat allocations included in the comparison.

The election-level outcome is the difference between these two seat shares, denoted by Δ_e and defined in Hypothesis 1. Positive values indicate that smaller parties collectively receive a larger seat share under Double Pukelsheim while negative values indicate a smaller share. A value of five percentage points therefore means that their combined share of the analyzed seats is five percentage points higher under Double Pukelsheim. This also means, that a negative or positive aggregate difference, does not mean that each included party gains or loses seats respectively.

Hypothesis 1 is assessed using the average seat-share differnce aross all 31 elections. In that analysis every elections receives equal weight, which means that cantons with more elections held under DP are influencing the average stronger. Since the parliament sizes differ, a supplementary weighted mean is added, where elections with a bigger parliament influence the mean stronger. It expresses the total net seat gain of smaller parties across the included elections as a percentage of the total analyzed seats, giving greater weight to elections with more seats. To better display the variation and typical gain, the median ,the first and third quartiles are reported additionally. The main summaries exclude Graubünden for the reasons set out in Section 6.2 its observations are presented separately for context.

All three exploratory analyses are descriptive. They examine differences across institutional settings and successive elections without isolating the causal effects of individual electoral rules.

For EQ1, election-level seat-share gains are first averaged within each canton. These canton-level outcomes are examined against mean district magnitude, median district magnitude, the share of districts below nine seats, and CV. These measure are all calculated from the same fixed allocation in each canton's last pre-reform election and the Schaffhausen parliament adjusted to 60 seats. These historical characteristics are held fixed when comparing the canton-level average gains across the analyzed elections. Uri covers the same four municipalities included in the HB–DP comparison, and Valais's district count refers to its 14 Hagenbach-Bischoff allocation districts. CV divides the population standard deviation of the baseline district magnitudes by their baseline mean. A higher CV indicates greater dispersion relative to average district size. All district measures are calculated once per canton from this baseline and are held fixed across its analyzed elections. For each measure, cantons are grouped below or at or above its unweighted mean across the eight transition cantons. Group means give each canton equal weight, and the group difference is calculated as below-mean minus at-or-above-mean. Graubünden is shown only for context and excluded from cutoffs and group means.

The historical reference year applies to all district measures. Historical seat allocations are documented in the cantonal publications and court judgments cited below.[4] Later district magnitudes may change, so the district measures describe associations with the initial structure. For Schaffhausen, the 2004 structure is adjusted from 80 to 60 seats using the published hypotethical allocation in the

[4]Sources by canton and reference year: Aargau 2005 (Kanton Aargau, 2004); Graubünden 2018 (Kanton Graubünden, n.d.); Nidwalden 2010 (BGE 136 I 352, E. 3.5); Schaffhausen 2004, adjusted to 60 seats (Regierungsrat des Kantons Schaffhausen, 2007, Table 1, p. 2); Schwyz 2012 (Urteil des Bundesgerichts 1C_407/2011 vom 19. März 2012, Sachverhalt A); Uri 2016 (Standeskanzlei Uri, 2016, p. 5); Valais 2013 (BGE 140 I 107, Sachverhalt A); Zug 2010 (Staatskanzlei des Kantons Zug, 2010, p. 92); Zurich 2003 (Regierungsrat des Kantons Zürich, 2002).

government's reform report (Regierungsrat des Kantons Schaffhausen, 2007, Table 1, p. 2). The election-specific Hagebach-Bischoff to Double Pukelsheim allocations are unchanged.

For EQ3, elections are grouped by whether or not an explicit electoral quorum applied under Double Pukelsheim. Within each group, the mean of Δ_e gives each election equal weight, so cantons with more observed elections contribute more. The group difference is calculated as the mean without a quorum minus the mean with one. Graubünden is shown for context and excluded from both means. Since the observations differ structurally in terms of quorum levels, geographical scope and others, the analysis is only descriptive.

To examine successive elections, the seat-share difference Δ_e is plotted separately for each canton and its successive elections. The aim is to analyze whether the mechanical seat-share gain persists across elections. To identify possible influencing factors, they are noted next to the election in which they ocurred. Later changes and fluctuations are interpreted with the help of the additional context provided by the notes. Besides the separate analyis of each canton, the results are also set into the between-canton context to possibly identify general patterns.

6.5 Results

6.5.1 Hypothesis 1: Smaller parties' seat-share gain

Figure 1 compares smaller parties' combined seat shares under the observed Double Pukelsheim allocation and the counterfactual Hagenbach-Bischoff allocation for every cleaned election. Smaller parties are defined as parties with less than 10% of the vote share in the analyzed area. The x-axis reports their combined share of analyzed seats, while the y-axis reports their combined vote share. The grey line connects the two allocations derived from the same election, and each election is labelled at its Double Pukelsheim point. This seat-share difference is the outcome used to assess Hypothesis 1 and allows comparisons across elections with different numbers of analyzed seats. The corresponding comparison in absolute seat counts is provided in Figure 7 in Appendix 3.

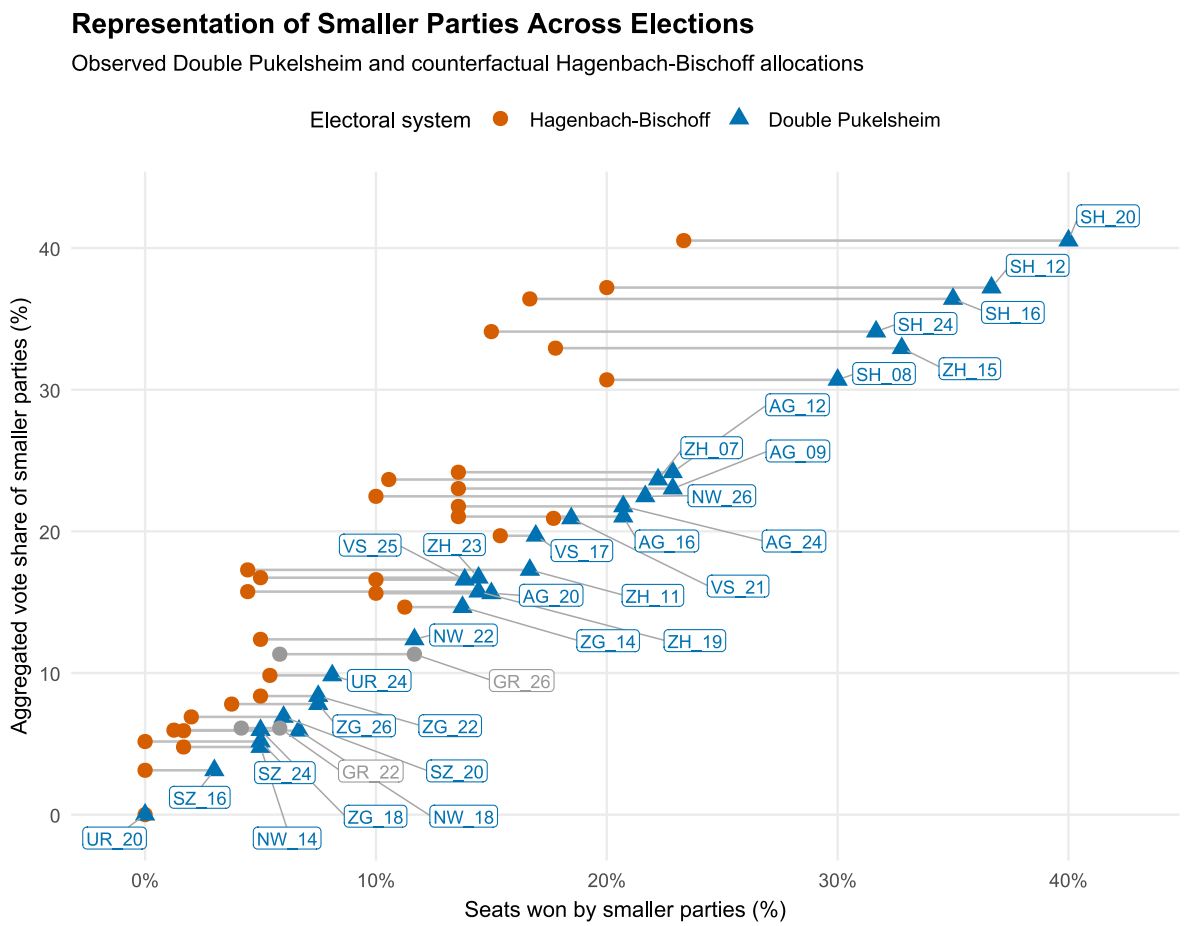


Figure 1: Smaller-party seat-share gain across all 31 elections

> **i** Hypothesis 1 summary
>
> Across all 31 elections (Graubünden excluded), the unweighted mean gain in smaller parties' share of analyzed seats was 7.6 percentage points (Q1: 3.5; median: 6.7; Q3: 10.8). The mean weighted by each election's number of analyzed seats was 7.8 percentage points. 16 individual elections exceeded five percentage points.

These results descriptively support Hypothesis 1 and indicate that smaller parties indeed in the 31 elections held receive a larger share of the analyzed seats under Double Pukelsheim than under Hagenbach-Bischoff. The effect is especially pointed out, through that in every election where small parties contested (every election but Uri 2020), they aggregately received a larger seat share. This finding is in accordance with Leuzinger's observation for the seven from him examined elections (Leuzinger, 2018, p.42). The weighted mean, unweighted mean and median are all relatively close to each other, indicating that the effect is not driven by a few elections with many seats and that the positive effect isn't determined by extreme outliers.

The average gain differs across the cantons for e.g. is there the Canton of Schaffhausen, with a stable high seat gain for smaller parties. There the high seat-share gain can be partially explained through the decision to treat affiliated parties as independent lists. Therefore these affiliated lists were classified as smaller parties, if they didn't reach the 10% threshold themselfs and with that higher the relative

gain for smaller parties. Across the five Schaffhausen elections, the average effect of this decision is that the relative gain drops from 15.67 to 8.67 percentage points. Taking the effect into account, the mechanical transition effect in Schaffhausen is still relatively big compared to other cantons. Cantons with constantly relatively low stable seat gain shares on the other hand are for e.g. Uri, Zug and Schwyz.

The general finding is further robust to alternative definitions of smaller parties. Using a vote-share cutoff of less than 8%, the mean gain is 6.77 percentage points. Also do the smaller parties c further smaller parties collectively receive a higher seat share under Double Pukelsheim in 28 of the 29 where atleast one parliamentary party with a maximum of 8% vote-share contested. The exception is VS 2021 , where the aggregated difference is zero and the upper-allocation process isn't done across the whole canton. With a cutoff of less than 12%, the mean gain is 7.91 percentage points. Regarding the collectively higher seat share in this case, in 30 of the 30 relevant elections, the smaller parties received a higher seat-share gain. With again Uri 2020, being the exclusion, where no party with less than 12% vote-share ran.

As a further sensitivity and robustness measure, the results were tested with a leave one canton out check. In that check each canton is omitted once and then the weighted and unweighted mean gain are recalculated. The positive seat-share gain persisted across each check. Which fits the finding that in 30 of 31 elections, there was a positive aggregated seat-share gain for smaller parties. The detailed results for the checks can be found in the Appendix in Section 10.4.

These measures deliver a profound picture over the aggregated seat-share gain of smaller parties, but they don't reveal which category of smaller parties profit most from the transition. Is it rather smaller parties between 3-4% or ones with 8-10%? Therefore this subanalysis examines this for all 31 transition elections. Both party-size figures additionally show observations from the two Graubünden elections for context. Since Graubünden previously used Majorz, its HB comparison is hypothetical and its results are only contextual.

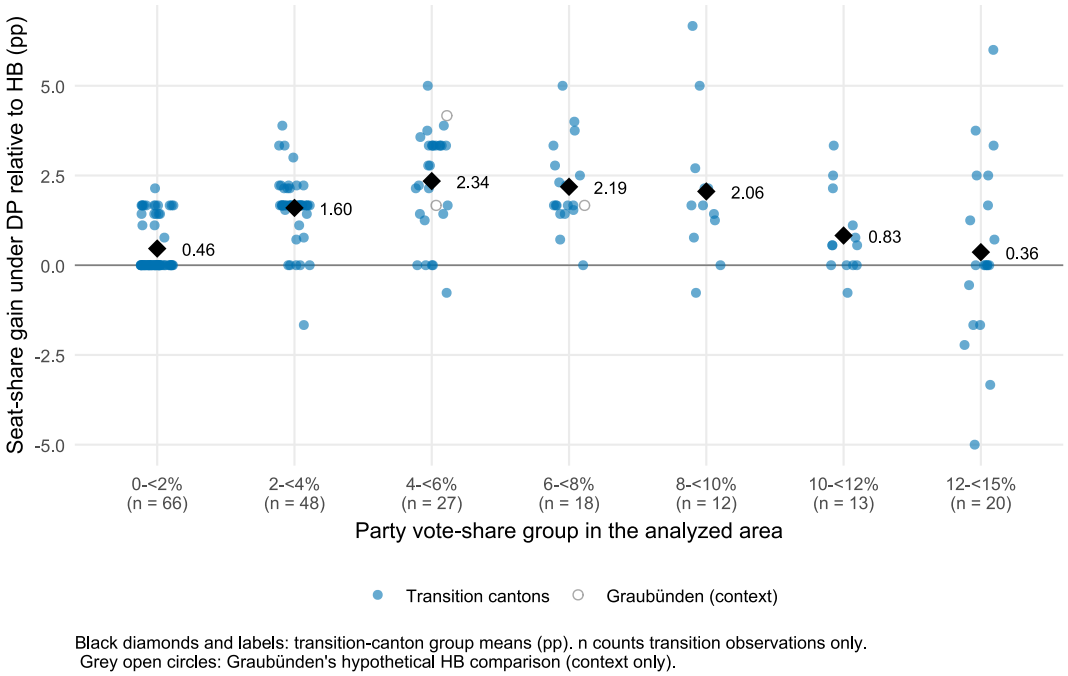


Figure 2: Mechanical seat-share gains by party vote-share group

The summary of the 204 party-election observations shows that parties between 4-6% vote share tend to profit the most from the implementation of the Double Pukelsheim. For parties with more than 10% the seat-share gain is on average markedly lower. For parties with atleast 12 and below 15% the average gain reaches a point, where the parties with 0.46 percentage points barely profit anymore from the implementation. On the other side of the scale for the parties below 2% one can see that they profit slightly, while there are multiple observations were a list didn't profit at all. From the 66 observations below 2%, 45 lists receive no seats under eiter allocation. This partly explains the limited gains for parties in that group. Overall does the analysis, show that smaller parties tend to profit from the implementation whereas bigger parties in average subsequently lose their prior advantage under the old system. To be able to identify a possible gradual pattern, the analysis was also done without grouping. The figure there shows no clear monotonic relationship, further can the substantial variation between parties of the same vote share be seen throughout the analysis. With the observation, that observations at a very low share cluster around zeero, while positive gains occur across most of the remaining range. Further do most negative observations appear in the upper end of the range. The figure can be found in the Appendix in Section 10.3.

To sum up the analysis of Hypothesis 1: It covers all 31 cantonal parliamentary elections held under Double Pukelsheim in Switzerland from the cantons that transitioned from Hagenbach-Bischoff. Across these elections, smaller parties received under all definitions of smaller parties, a higher aggregate seat share under Double Pukelsheim. This pattern also persisted throghout almost every within election comparison of smaller parties. Further did the analyis show, that parties between 4% and below 10% profited most from the implementation of the Double Pukelsheim. These findings are based on the mechanical allocation difference conditional on the observed votes and specified counterfactual rules and therefore do not capture the whole transition effect. The next section examines whether the magnitude of this effect is associated with the cantons district magnitudes.

6.5.2 Exploratory question 1: Differences by district magnitude

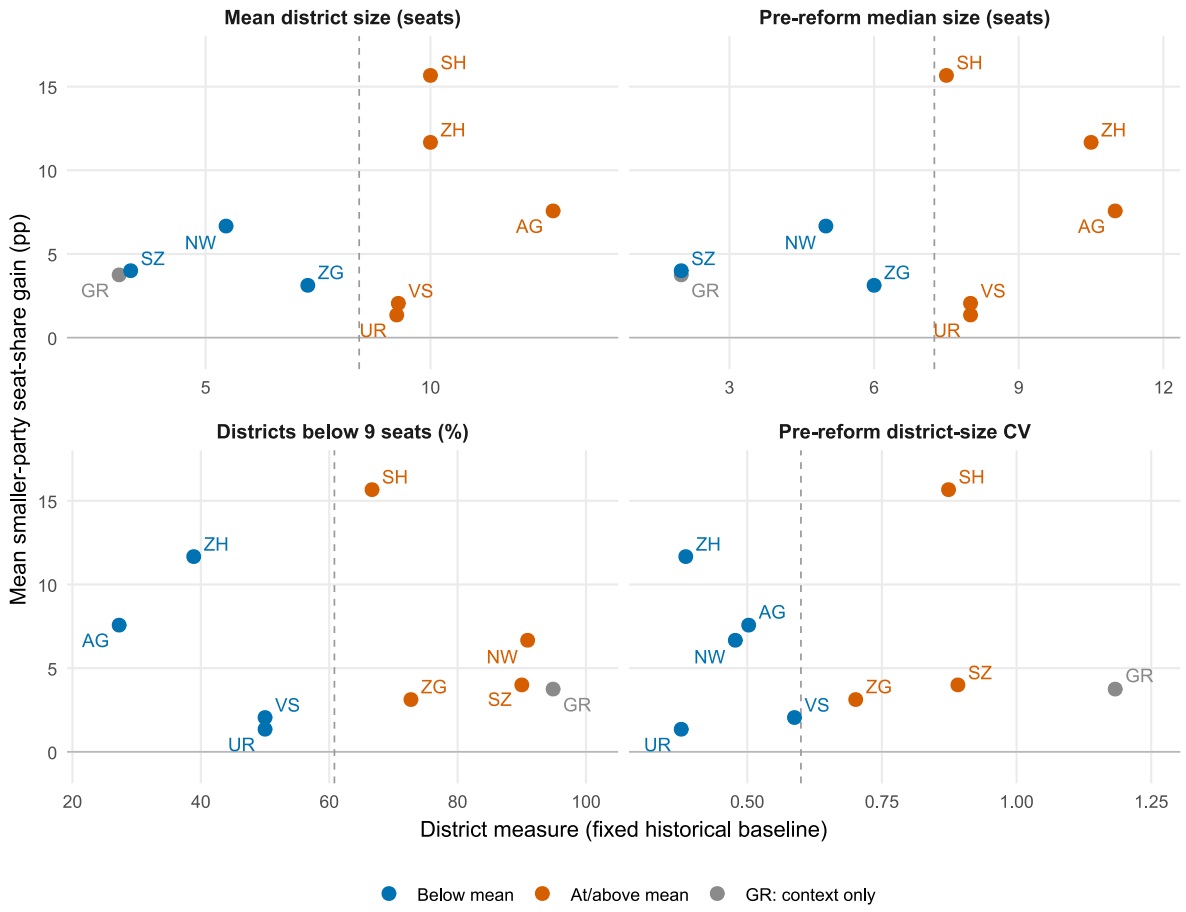
This section examines how district magnitude varies across the cantons included in the analysis, considering both typical district size and variation within cantons. Table 2 summarizes these differences using district counts, the share of districts with fewer than nine seats, mean and median district magnitudes aswell as the coefficient of variation. All measures refer to the fixed historical baseline defined in Section 6.4 , with Schaffhausen adjusted to 60 seats. Graubünden is shown for context only.

Table 2: Pre-reform district structure by canton

Canton	Districts	Reference year	Mean seats	Pre-reform median	Below 9 seats (%)	Pre-reform CV
AG	11	2005	12.73	11.00	27.3	0.50
GR	39	2018	3.08	2.00	94.9	1.18
NW	11	2010	5.45	5.00	90.9	0.48
SH	6	2004	10.00	7.50	66.7	0.87
SZ	30	2012	3.33	2.00	90.0	0.89
UR	4	2016	9.25	8.00	50.0	0.38
VS	14	2013	9.29	8.00	50.0	0.59
ZG	11	2010	7.27	6.00	72.7	0.70
ZH	18	2003	10.00	10.50	38.9	0.39

Notes Source: author's calculations using official cantonal allocations and court judgments. To compute the measures for Uri, only the four DP consitutiencies (Municpialities) were taken.

Figure 3 relates each district measure to the canton-level mean gain in smaller parties' seat share. Colours distinguish cantons below and at or above the unweighted cross-canton mean of the respective measure. The cutoffs are defined by the mean of the eight relevant cantons and are the following: 8.42 seats for the unweighted mean magnitude, 7.25 seats for median magnitude, 60.81% for the share below nine seats, and 0.60 for the CV. A CV of 0.6 means, that the standard deviation is 60% of the mean. The cross-canton means are marked with dashed lines.



Dashed lines: means across eight transition cantons. GR is contextual; Uri covers four DP municipalities.

Figure 3: Smaller-party seat-share gains by district magnitude and dispersion

Figure 2, shows no clear, consistent relationship between the former district structure and the relative smaller parties seat gain across the eight relevant cantons. For the mean and median district magnitudes, it was expected that cantons with smaller districts would see larger gains for smaller parties. Instead is the analysis slightly pointing into the opposite direction, as also visible in the later discussed Figure 4. Moreover is no clear pattern visible when comparing the groups above and below the 60.81% share of districts with a maximum of eight seats. The observation of a missing clear correlation, is further the case in the analysis on the pre-reform district-size cv. Followingly in Figure 4 , the same outcomes will be displayed as grouped dot plots with group means which makes the differences between the groups visually easier to assess. The missing paterns may be caused by the restricted amount of cantons and their different application of the Double Pukelsheim.

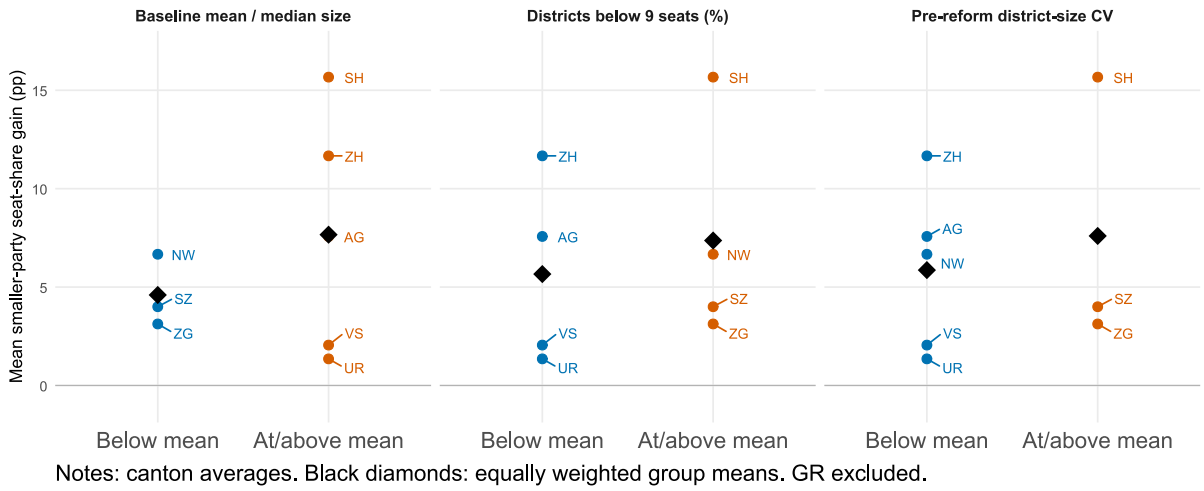


Figure 4: Smaller-party seat-share gains by historical district characteristics

Notes: Cantons are grouped below or at or above the cross-canton mean of each historical district measure.

For both mean district magnitude and pre-reform median district magnitude, the average gains for the below mean group are 4.60 and 7.66 percentage points for the at/above mean group. The at/above group results, are relatively spread out, while the below groups cantons are quite close to eachother. The below-minus-above difference is −3.06 percentage points and with that opposite to the expected direction.

For the share of districts below nine seats, the average gains for the below mean group are 5.66 and 7.36 percentage points for the at/above mean group. The higher-share group gains of 1.70 percentage points more are consistent with the expected direction. However is in the below group Schaffhausen included, whose comparatively high gain raises the group average. As discussed in Section 6.5.1 does the separate treatment of affiliated lists contribute to Schaffhausen's high gain. Therefore the result needs to be interpreted sensitively.

For pre-reform district-size variation, the average gains for the below mean group are 5.86 and 7.60 percentage points for the at/above mean group. The below-minus-above difference is therefore −1.74 percentage points. This means that the group of cantons with the higher variation in district sizes relative to their mean in average had higher gains. Here, too Schaffhausen's relatively high gain raises the group average, leading to that the result should be interpreted sensitively.

To analyze the sensitivity of the results to certain cantons, a leave one canton out check was carried out. The check showed that the results in exploratory question 1 are sensitive to the omission of the canton of Schaffhausen. The omission of the canton of Schaffhausen, lets the mean and median measure decline from −3.06 to −1.06. For the share below 9 seats and the CV measure though the direction switches with the omission of the canton. This means that without the canton of Schaffhausen, the average gains for the below mean groups are higher than for the above mean group, which oposses the estimated direction. Generally is there substantial variation in both groups, which means that also the omission of cantons like Uri and Valais have substantial impact on some of the measures. The exact results to the robustness checks can be found in Section 10.4.

Taken together, the four district measures reveal no consistent pattern linking the initial district structure to smaller parties' gains. This is combined with substantial variation within the analyzed groups. Further are the four measures sensitive to the omission of single cantons. Which supports the assesment of a missing pattern. The sensitivity to Schaffhausen can be linked to its comparatively high seat-share gain. Which is partly caused as discussed in Section 6.3 by the separate classification of affiliated lists as smaller parties. These descriptive comparisons therefore can not show any causal effect of district magnitude or its dispersion.

6.5.3 Exploratory question 2: Explicit electoral quorums

Exploratory question 3 researches on whether the implementation of explicit electoral quroums have an influence on the smaller parties seat-share gain. The explicit quroums ensure that only parties which reach a specific vote-share are considered for the allocation of seats. This may prevent smaller parties from entering the parliament, which is why it's expected in this thesis, that in cantons with a quorum, the seat-share gain of smaller parties is reduced. This directional estimation is analyzed followingly:

Table 3: Situation regarding explicit electoral thresholds under Double Pukelsheim in the elections studied.

Canton	implemented Thresholds	relevant years
Aargau	none	2009
Aargau	5% district level or 3% cantonal level	2012, 2016, 2020, 2024
Graubünden	3% cantonal level	2022, 2026
Nidwalden	none	2014, 2018, 2022, 2026
Schaffhausen	none	2008, 2012, 2016, 2020, 2024
Schwyz	1% cantonal level	2016, 2020, 2024
Uri	none	2020, 2024
Valais	8% district level	2017, 2021 ,2025
Zug	5% district level or 3% cantonal level	2014, 2018, 2022, 2026
Zurich	5% district level	2007, 2011, 2015, 2019
Zurich	5% district level or 3% cantonal level	2023

Source: Author's compilation. Threshold rules are based on the sources discussed in Section 4.1.4. Election years refer to the elections included in the empirical analysis, whose data sources are documented in Section 10.1.

As documented in the table, out of eight analysis cantons, five applied a threshold. With Aargau being a special case, since the canton applied it only after their first election under Double Pukelsheim. The most often used explicit qourum is the combination of a 5% district-level and a 3% cantonal level threshold of which atleast one has to be reached. This combination was used in 9 of 19 elections held with an explicit threshold.

The difference is visualized followingly, in a bivariate analyisis presented as a grouped dot plot.

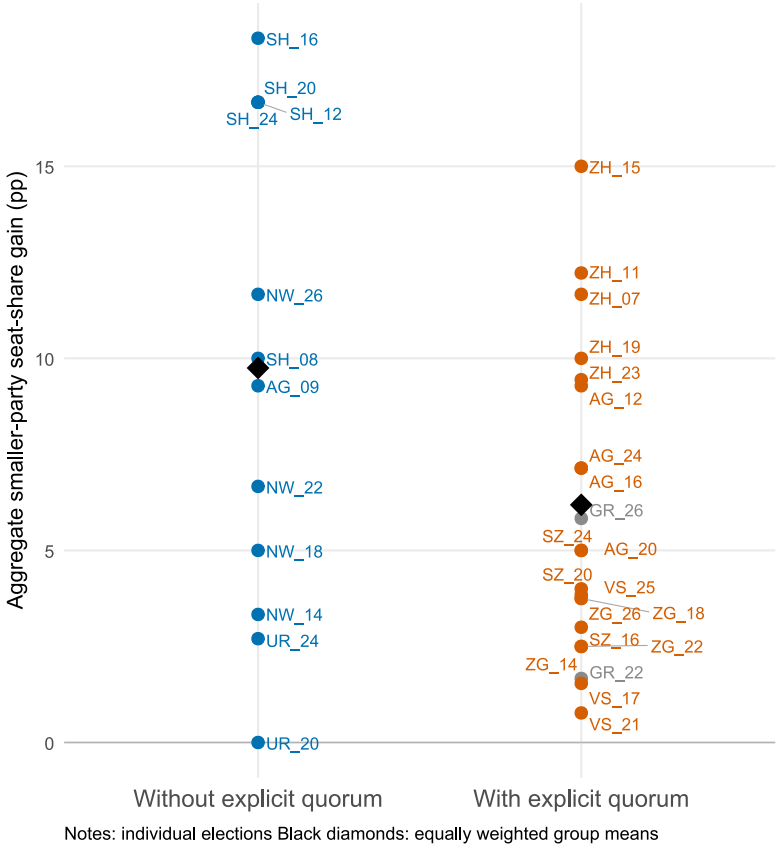


Figure 5: Smaller-party seat-share gains by explicit electoral quorum.

Figure 5 provides a descriptive comparison of smaller-party seat-share gains by the presence of an explicit electoral quorum. Average gains are higher in the group without an explicit quorum (9.7 percentage points) than in the group with one (6.2 percentage points). Both groups exhibit substantial variation. From which the canton of Schaffhausen sticks out, as mentioned in prior analysises.

To analyze whether the effect holds despite the variation, the leave one canton out sensitivity check was performed. As priorly is the high seat-share gain of Schaffhausen making the analysis and the results sensitive to the omission of it. Schaffhausen accounts for five of the twelve elections in the group without an explicit quorum, with all five recording seat-share gains above the group average. With Schaffhausen's exclusion is the difference flipping from 3.56 to −0.67. That means without Schaffhausen, the observed effect is not aligned anymore with the expectation, that in cantons without an explicit quorum the seat-share gain is higher. The exact results for the omission of every canton are outlined in Section 10.4.

Especially interesting, is the case of the canton Aargau, where in 2009 the election was held without a threshold and in the elections 2012, 2016, 2020 and 2024 with a threshold. The data shows that the seat-share gain for the pre-reform election 2009 and the first post-reform election 2012 is equally big with 9.3 percentage points. In the subsequent elections, though the seat-share gain droped accordingly to the assumption. While the unchanged gain between 2009 and 2012 contrasts with the assumption, the subsequent elections are inline with it. For Aargau there is therefore no clear pattern visible, moreover might the seat-share gain difference be influenced by further unobserved circumstances.

To summarize the analysis about the effect of the implementation of explicit quoroms, it can be stated that within the 31 relevant election a difference which matches the expectaion can be seen. This difference though is sensitive to the inclusion of Schaffhausen. Further is the limited amount of elections held till date and the different application of thresholds and electoral law, encouraging to interpret the difference with caution.

6.5.4 Exploratory question 4: Change across successive elections

The preceding analysis showed that cantons differ in their implementation of the Double Pukelsheimer system. These institutional differences motivate a closer examination of within-canton patterns. The last exploratory question therefore examines how the smaller parties seat-share gain changed across successive elections within a canton. For that each canton and it's successive changes are analyzed separately. To possibly better understand changes, events such as electoral reforms and changes in the party composition are documented. Generally stable positive values of Δ_e indicate a persistent mechanical advantage for smaller parties, while substantial fluctuation or even changes in sign would suggest that the effect is strongly dependent on the individual circumstances of an election.

Since Figure 6 aligns every Double Pukelsheim election for each canton independently by their order, the first election under the system is numbered 1, the next is numbered 2, and so on. This means that the same election number represents different calendar years across cantons. As in the main hypothesis the outcome is the observed Double Pukelsheim seat share minus the counterfactual Hagenbach-Bischoff seat share, expressed in percentage points of the seats included in the comparison. The smaller-party category is recalculated for each election using the existing vote-share cutoff of less than 10%.

Regarding the events and their noteboxes, there are two different formats. The bigger boxes mark electoral rule changes, from which there were three relevant ones: Aargau's introduction of a 5% district-level or 3% cantonwide quorum in 2012 (second election); Schaffhausen's majority clause, applied in 2016 (third election); and Zurich's additional 3% cantonwide eligibility route, first applied in 2023 (fifth election). The Zurich rule supplements the existing 5% district-level quroum and as in Aargau, the parties have to only reach either 5% within a district or 3% on the cantonal level. These dates follow the institutional history in Section 4.1.4; Schaffhausen's amendment history records 1 March 2016 as the effective date of implementation (Kanton Schaffhausen, 1904/2019). The E/L annotations mark parliamentary parties entering or leaving the below-10% vote-share group; a preceding 2 indicates two parties. Graubünden is shown as priorly only contextual.

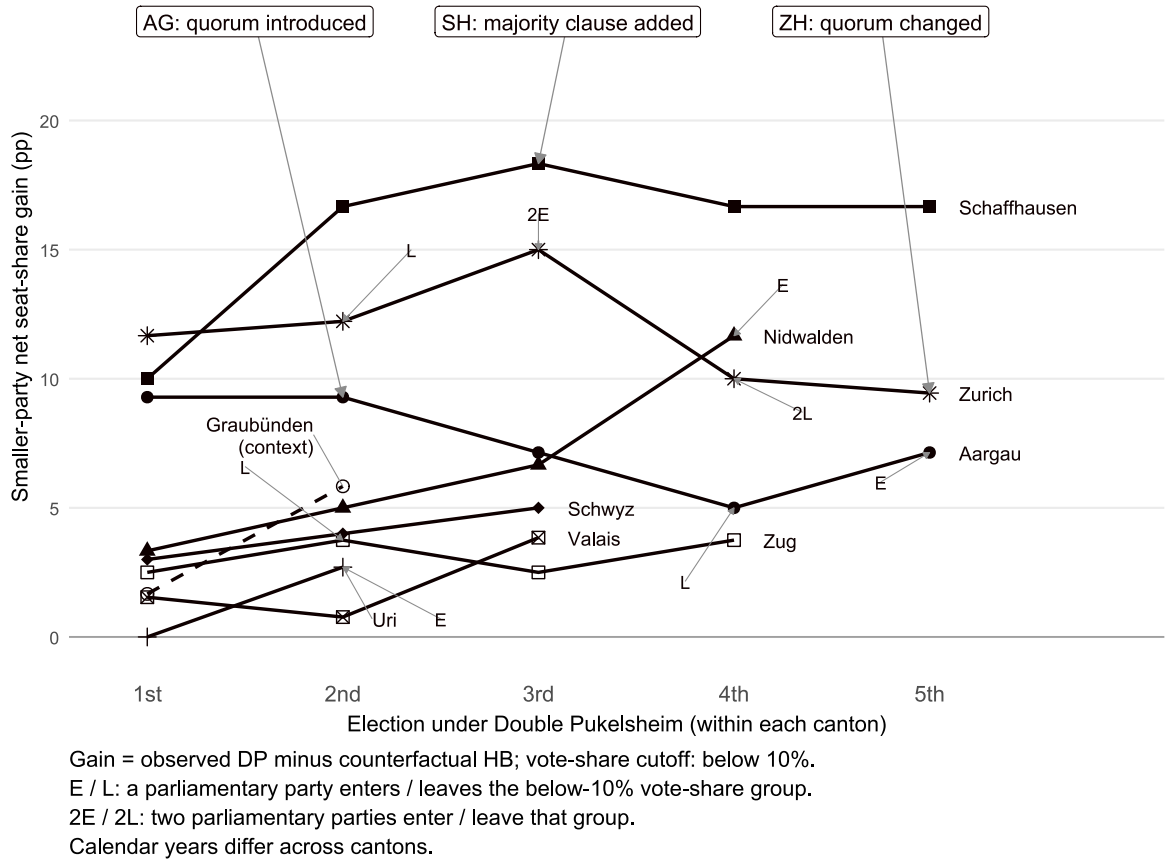


Figure 6: Smaller-party seat-share gains across successive elections under Double Pukelsheim

Notes Sources: Large text boxes are based on the sources discussed in Section 4.1.4.

i EQ4 summary

Excluding Graubünden, the average within-canton change in smaller parties' seat-share gain was +1.6 percentage points from election 1 to 2 (8 cantons); +1.0 percentage points from election 2 to 3 (7 cantons); −0.5 percentage points from election 3 to 4 (5 cantons); +0.5 percentage points from election 4 to 5 (3 cantons). The average seat-share gains across the cantons were 5.2 percentage points in election 1 (8 cantons); 6.8 percentage points in election 2 (8 cantons); 8.4 percentage points in election 3 (7 cantons); 9.4 percentage points in election 4 (5 cantons); 11.1 percentage points in election 5 (3 cantons). Each average gives equal weight to the cantons observed in both elections of the respective transition.

Generally spoken, are the seat-share gains with some canton-specific fluctuations, relatively persistent. As seen in the analysis of Hypothesis 1, is with Uri 2020 only one election, in which there was no positive seat-share gain. This is the case, since there simply no *smaller_party* ran. Another interesting observation, is the fact that the average seat-share gain rises successively from election to election. This with the caveat of the limited number of cantons and that this number lowers throughout the later elections. For a possible effect speaking is, that the cantons of Schaffhausen and Zurich which appear in all comparisons and have relatively high seat-gain shares also experience drops in their

share throughout the elections. This while the average seat-share gain rises continously. Additionally is the analysis, less dependent on political trends specific to a single election year, since for example the second elections held under Double Pukelsheim range from 2011 in Zurich to 2024 in Uri. This though does not isolate the pattern from political trends. A possible partial explanation for the observed pattern, is the broader increase in party-system fragmentation, which was documented by Lüscher (2024, pp. 133-135). This because if smaller parties receive wider support, their combined disadvantage under the former Hagenbach-Bischoff and with that their seat-share gain under Double Pukelsheim may increase. Nevertheless, whether this mechanism actually explains the observed pattern remains untested.

Besides the broader image, there are also interesting patterns within the cantons. The biggest changes between two elections, can be seen for Schaffhausen, Zurich and Nidwalden. In Schaffhausen the seat-share gain rose from the first to the 2nd election from roughly 10 to roughly 17 percentage points, where it settled afterward quite stable. A possible partial reason, is the higher number of smaller parties in the parliament which increased from seven in 2008 to nine in 2012. From the 2015 Zurich, to the 2019 Zurich election, the seat-share gain dropped by roughly 5 percentage points, a reason for this is possibly, that both the GLP and the Greens exceeded the 10% smaller-parties threshold while the BDP lost its parliamentary representation. This led to the circumstance, that from seven smaller parliamentary parties in 2015, the number dropped to four in 2019. In Nidwalden since the implementation the seat-gain share rose continously. The strongest difference was observed from the 3rd to the 4th election, where the share rose by 5 percentage points. This effect might be partially influenced by the in the smaller party composition, since the green party fell below the 10% threshold. For Aargau, Zug, Schwyz and Valais we can observe relatively stable patterns with smaller fluctuations. For the canton of Uri only two elections were observed, which is why no pattern can be inferred.

In the three cantons Aargau, Schaffhausen and Zurich, the Double Pukelsheim system, was adjusted after its implementation. The case of the canton of Aargau was priorly discussed in Section 6.5.3. The assumption of the implementation of a threshold is that the smaller-party seat share gain declines with it. The post electoral change results for the canton of Aargau don't present a clear picture there with results aswell in accordance as also speaking against the assumption. The absence of a clear effect, may be partly explained through that in each election the biggest list without parliamentary representation, received only between 0.1% to 0.7% vote shares.[5] This suggests that the explicit quroums in Aargau only had limited scope to exclude parties, which otherwise would have won seats. In Schaffhausen in 2016 as described in Section 4.1.4, a majority clause was implemented. This majority clause ensures that the strongest list in each constituency receives at least one seat. It does this, while preserving the seat allocation of the canton-wide upper apportionment. (Kanton Schaffhausen, 1904/2019, Art. 2d paras. 1bis and 2(b)) Thus it has no effect on the smaller party seat-share gain. In the canton of Zurich, in the year 2023 a quroum change applied. Since to the existing 5% district-level quroum a 3% canton-wide one was added, of which parties have to reach either one. This electoral change is in the tendency highering the seat-share gain, since it possibly enables more parties to reach the seat-allocation process. In the 2023 election, though no party profited from this clause, since every qualifying party also reached the 5% district-level threshold (Regierungsrat des Kantons Zürich, 2023,

[5]For cantonal vote shares and parliamentary representation, see T. Schaub (2012, p. 12), Staatskanzlei des Kantons Aargau (2016b, p. 2), Staatskanzlei des Kantons Aargau (2020, p. 2), and Staatskanzlei des Kantons Aargau (2024).

p.9). Further did the seat-share gain drop from the 2019 to the 2023 election. Overall did the three electoral changes therefore have no visible effect on the smaller party seat-share gain.

As for the other analyses, was here also a robustness check performed. The check was done for each election number separately. Throghout all checks does the seat-share gain remain positive. The mean change though from election three to four and from four to five switches signs when Zurich, respectively the canton of Aargau is excluded. These later transition include only five respectively three cantons and are therefore more sensitive to indidvidual cantons. The exact statistics for the robustness check can be found in Section 10.4.

The analyis across all 31 elections, therefore indicates that the mechanical seat-share gain for smaller parties persists across sucessive elections. The strength and variation in the magnitude differ between the cantons. In the analyzed elections, the gain not only persisted, but rose across sucessive elections. The average within-canton change though isn't rising consecutively with a small decline between the third and fourth election. Moreover is the analysis for the within-canton change sensitive to the canton of Zurich, respectively Aargau in the two last transitions. The analysis and it's effects may be partly affected through the changes in the party composition of the smaller parties group. The results therefore imply a consistent gain for smaller-parties across cantons and elections. However it can't be established and is not inferred or mentioned by the literature, that additional elections under Double Pukelsheim increase the gain.

7 Discussion

In this paper, the general question: "20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?" was answered in two parts. In the first step, the mechanical effect on the smaller parties was examined. Subsequently the study examined which conditions could lead to differences in the strength of the mechanical effect across cantons or elections.

Regarding the first step, it was found that across elections the smaller sparties collectively gain seat shares under Double Pukelsheim compared to Hagenbach Bischoff. The average aggregate unweighted seat-share gain was 7.6 percentage points, further did in 30 of the 31 analyzed elections, the smaller parties aggregately receive a higher seat-share. In the Uri 2020 election, no eligible smaller party contested, thus no seat-share difference could be observed. The gain also persisted when adapting the 10% smaller parties threshold to 8% and 12%, and after the exclusion of any single canton. Further was it observed, that out of the smaller parties group, the lists with atleast 4% and below 6% across all elections profited the most from the transition. The findings in Hypothesis 1 are consistent with the idea of the upper apportionment process. It lowers the share of ineffective votes and with that highers the equality of electoral sucess. (Lüscher, 2024, p. 3)

This finding matches the observations of the existing literature. Leuzinger states that in his analysis across seven elections, smaller parties generally received more seats under the new system (Leuzinger, 2018, p.42) Also in the by Pukelsheim and Schuchmacher analyzed Zurich 2011 election and in the Schwyz 2016 election by Lüscher, did smaller parties profit from the implementation (Pukelsheiim & Schuhmacher, 2011 ; Lüscher , 2024, p.13).

In the second step, potential differences in the effect according to the different implementation of the Double Pukelsheim were examined. First it was analyzed with three measures whether the pre-reform district sizes have an impact on the seat-share gain or not. No consistent relationship between the prior constituency sizes and the gain for smaller parties was found. This is contrary to the formulated expectation. The expectation was, that in cantons with smaller pre-reform districts, were the high natural thresholds impose a greater disadvantage, that there the upper apportionment and it's cantonwide mechanism would have a greater effect. The expectation, was also based on Lüscher's finding that in cantons which priorly had smaller districts, the proportionality had increased stronger (Lüscher, 2024, pp.132-133). Thus that the analysis here finds no evidence, is it not contradicting Lüscher's findings. Since an improvement in overall proportionality doesn't need to have a direct impact on the seat-share gain for smaller parties.

A sedond exploratory question, examined whether there is a difference in the mechanical seat-share difference between elections held with and without an explicit quroum. The examination here observed a higher seat-share gain for cantons without an explicit quroum. This fits the literature, since Lüscher and the Federal State Chancellery, both argue that explicit threshold can restrict the gains among smaller parties. Through which the greater proportionality achieved through the upper apportionment gets partly offsetted (Bundeskanzlei, 2013, p. 18; Lüscher, 2024, pp. 107, 135). Thus of the seen effect in the analyis and the support of the theory for an effect, is the effect seen in this thesis analysis sensitive to the inclusion of Schaffhausen. This to a point, that without the Schaffhausen elections, the effect reverses. Therefore the observed effect needs to be interpreted cautiously.

The third exploratory question analyzes the seat-share gains across sucessive elections within and with respect to each cantons electoral rules. Through this approach, it brings together elements of the preceding analyses. The results show that the positive seat-share advantage generally persists across successive elections, further supporting the main finding. Although average gains are higher at later election numbers, mean within-canton changes do not follow a consistently increasing pattern. Moreover, fewer cantons contribute to the later comparisons, and some changes are sensitive to the inclusion of individual cantons. The findings therefore do not establish that additional elections under Double Pukelsheim increase smaller parties mechanical seat-share gain.

In order to analyze all relevant elections and their various underlying assumptions, a few assumption had to be made. The analyis, holds all post-reform votes constant and identifies the conditional mechanical difference, without capturing the behavioural effect of the transition. For the cantons, in which priorly list alliances were allowed, are in this thesis no hypothetical list alliances formed. This decision might in the relevant cantons understate the smaller lists representation under the Hagenbach-Bischoff counterfactual. Further was each list, which ran independently, treated as a separate party, since the Double Pukelsheim algorithm also treats them separately. This decision is especially relevant in Schaffhausen, were across multiple elections, several affiliated list such as Youth or special-interest lists ran. In addition to this decisions, do the limited amount of eight cantons and their different electoral laws, limit the extent to which the observed differences in gains can be attributed to a single feature.

For cantons considering a transition from Hagenbach–Bischoff to Double Pukelsheim, the following can be stated. With 30 of the 31 analyzed elections showing a collective seat-share gain for smaller

parties, the findings suggest that a similar mechanical advantage may arise in other cantons adopting DP. However, its magnitude depends on canton-specific circumstances, such as electoral rules, district sizes, and vote distributions. Behavioural responses by parties and voters introduce further uncertainty. It therefore makes sense for cantons to calculate a hypothetical DP allocation based on their observed Hagenbach–Bischoff election results. This allows them to assess the potential mechanical seat-share changes while explicitly accounting for their electoral structure and the proposed DP rules.

Further research could extend this thesis in the following ways. With examining how plausible counterfactual list-alliances affect the estimated gains. As also with adressing the explicit quroums and their possible effects. Specifically it could recalculate each observed election under different electoral quroums, ceteris paribus, to isolate their mechanical effect on smaller parties representation.

8 Conclusion

In this paper, I examined the mechanical effect of the transition from the Hagenbach-Bischoff Method to the Double Pukelsheim on the seat distribution of smaller parties. The thesis focused on examining the seat-share differences between the two systems, for parties with less than 10% vote share. For this all 31 elections held by the eight transition cantons from HB to DP were, while holding observed votes constant analyzed.

For all held elections, the average unweighted seat-share gain was 7.6 percentage points for the aggregate smaller parties group. In 30 of 31 elections, did the smaller parties collectively gain seat-shares. With the Uri 2020 election being the exception, where no list below the 10% smaller parties threshold contested, thus no seat-share difference could be observed. The result of a positive seat-share difference, further persisted with alternative cutoffs for the smaller-parties group and when excluding individual cantons. This findings therefore descriptively support Hypothesis 1.

The further analysis, explored potential differences between cantons, based on pre-reform district sizes and explicit quroums. For the relationship between the pre-reform district sizes and the seat-share gain, no consistent pattern could be observed. Regarding differences between elections held with and without a quroum, a higher seat-share gain for elections without a quroum was observed. However is the observed difference sensitive to individual cantons. Further was it found, that average gains persisted and even increased across succesive election numbers, even when individual cantons were excluded. However were the average within-canton changes less consistent and sensitive to canton exclusions. This pattern doesn't establish that additional elections under Double Pukelsheim increase the seat-share gain.

However is this analysis only focusing on mechanical differences conditional on the canton specific rules and on the observed post-reform votes and not estimating the reforms complete effect. This paper extends the existing knowledge about the mechanical seat-share gains for smaller parties by covering all 31 available elections for the eight transition cantons, including the succesive elections within each canton. This enables a wider basis to assess the magnitude and consistency of the smaller parties advantage under the Double Pukelsheim.

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10 Appendix

10.1 Appendix 1: Explanation Data Cleaning Process

The table below identifies the election data and legal sources used in the canton-specific cleaning scripts and counterfactual Hagenbach–Bischoff allocation. It specifies source titles, election coverage, file types, and the purpose of publications used to reconstruct or validate the results, together with the relevant legal provisions. Clickable citations and source titles lead to online sources where available; unpublished material supplied by email is identified explicitly. Author–year and case citations correspond to entries in the bibliography.

Canton	**Election data: description and source**	**HB allocation: legal basis and source**
Aargau	*Grossratswahlen: Detailresultate, ab 1985*: official district results for 2009, 2012, 2016, 2020, and 2024 (Statistisches Amt Kanton Aargau, n.d.). *Detailergebnisse der Grossratswahlen vom 8. März 2009* and *Jahresbericht mit Jahresrechnung 2012*: validation of cantonal seat totals (Department für Finanzen und Ressourcen des Kantons Aargau, 2009); (Regierungsrat des Kantons Aargau, 2013). *Gemeinsame Grossrats- und Regierungsratswahlen: Vorverfahren gestartet* (2012) and *Ausschreibung der Grossrats- und Regierungsratswahlen 2016*: district seat magnitudes (Staatskanzlei des Kantons Aargau, 2012; 2016a).	*Grossratswahlgesetz (SAR 152.100), version in force from 1 November 2004*: initial allocation and allocation of remaining seats (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)).
Graubünden	*Pressedaten Proporzwahl kantonal: Grossratswahlen*: official election-result Excel workbooks for 2022 and 2026 (Kanton Graubünden, 2022; 2026).	*Grossratswahlgesetz (SAR 152.100), version in force from 1 November 2004*: basis for the hypothetical HB allocation (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)). *Verfassung des Kantons Graubünden (BR 110.100), version of 1 January 2018*: confirms the former Majorz system (Kanton Graubünden, 2003/2018, Art. 27).
Nidwalden	*Wahlergebnis der Landratswahlen*: unpublished Excel workbooks for 2014, 2018, 2022, and 2026, supplied by the Kanzleisekretariat via email (Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026).	*BGE 136 I 352*: Federal Supreme Court description of the former district-level HB allocation (BGE 136 I 352, E. 3.2).

Canton	Election data: description and source	HB allocation: legal basis and source
Schaffhausen	*Frühere Urnengänge (seit 2003)*: official result PDFs for 2008, 2012, and 2016 (Kanton Schaffhausen, n.d.). *Resultate der Kantonsratswahl*: official result PDFs for 2020 and 2024 (Staatskanzlei Kanton Schaffhausen, 2020; 2024).	*Proporzwahlverordnung vom 13. November 1979 (Amtsdruckschrift 2734)*: single-seat district rule, HB distribution number, and tie-breaking provisions. Original text supplied by the Staatsarchiv via email (Regierungsrat des Kantons Schaffhausen, 1979, §§ 2, 36–37).
Schwyz	*Kantonsratswahlen 2016: Gemeindeergebnisse*: 30 unpublished municipality-level CSV files supplied via email (Staatskanzlei Kanton Schwyz, 2016). *Kandidierendenresultate: Kantonsratswahlen 2020*: candidate-result CSV export (Staatskanzlei Kanton Schwyz, 2020). *Kantonsprotokoll der Erneuerungswahl Kantonsrat 2024*: election-result Excel workbook (Staatskanzlei Kanton Schwyz, 2024).	*Urteil des Bundesgerichts vom 19. März 2012 (1C_407/2011, 1C_445/2011 und 1C_447/2011)*: description of the former municipality-level HB allocation (Bundesgericht, 19. März 2012, E. 5.2).
Uri	*AuswertungKapitel2-2024*: unpublished Excel workbook containing the 2020 and 2024 results, supplied via email (Standeskanzlei Uri, 2024a). *Landratswahlen 2024: Statistische Auswertung*: official report used for party-label validation (Standeskanzlei Uri, 2024b).	*Synoptische Darstellung zum Gesetz über die Verhältniswahl des Landrats (Proporzgesetz)*: former allocation and tie-breaking rules in the column "Bisher" (Kanton Uri, 2018, Arts. 22–23).
Valais	Official district-level Grand Council result workbooks for 2017, 2021, and 2025 (Excel files). *Grossräte 2017 – Resultate: Offizielle Endresultate*: validation of the 2017 results (Staat Wallis, 2017). Grand Council validation-commission reports: Bericht zur konstituierenden Session vom 19. April 2021 and Rapport relatif à la session constitutive du 14 avril 2025; validation of cantonal vote and seat totals. *Verteilung der Anzahl Grossratssitze pro Bezirk* (legislatures 2017–2021 and 2021–2025) and *Anhang zur Verteilung der Anzahl Grossratssitze pro Bezirk, Legislaturperiode 2025–2029*: district seat magnitudes (Kanton Wallis, 2016; 2020; 2024).	*Gesetz über die politischen Rechte (kGPR; SGS 160.1), version of 9 April 2017*: statutory eligibility threshold and district-level HB allocation (Kanton Wallis, 2004a, Arts. 154–156).
Zug	*Kantonsratswahl Kanton Zug* (2014), *Kantonsratswahl 2018*, *Kantonsratswahl 2022*, and *Erneuerungswahl für den Kantonsrat 2026*: official open-data candidate-result CSV exports (Kanton Zug, 2022b; 2022a; 2022c; 2026).	*Wahl- und Abstimmungsgesetz (WAG), original text of 28 September 2006*: former HB and Majorz provisions (Kanton Zug, 2006, §§ 46–47, 55–56). *BGE 136 I 376*: confirms Neuheim's former Majorz system; Neuheim is included using a hypothetical HB allocation (BGE 136 I 376, E. 5.1).
Zürich	*Archiv Wahlresultate Kantonsratswahlen*: official party-result and candidate-result Excel workbooks for 2007, 2011, 2015, 2019, and 2023 (Amt für Statistik und Daten Kanton Zürich, 2026).	*Gesetz über die Wahlen und Abstimmungen (Wahlgesetz; LS 161), version of 1 January 2005*: former HB allocation and tie-breaking provisions (Kanton Zürich, 1983/2005, §§ 87–88b).

Table 4: Election data sources and legal bases for the counterfactual Hagenbach–Bischoff allocation.

10.2 Appendix 2: Code for Data Cleaning

10.3 Appendix 3: Additional Analysis

10.3.1 Smaller parties' absolute seat counts

Figure 7 supplements the seat-share comparison used to assess Hypothesis 1 in Figure 1. It shows the absolute number of seats won collectively by parties below the 10% vote-share threshold under the observed Double Pukelsheim allocation and the counterfactual Hagenbach-Bischoff allocation. The grey line connects the two outcomes for each election, with each election labelled at its Double Pukelsheim point. Because the number of analyzed seats differs across elections, these absolute counts provide context for the magnitude of the gains, while Hypothesis 1 is assessed using the difference in seat shares.

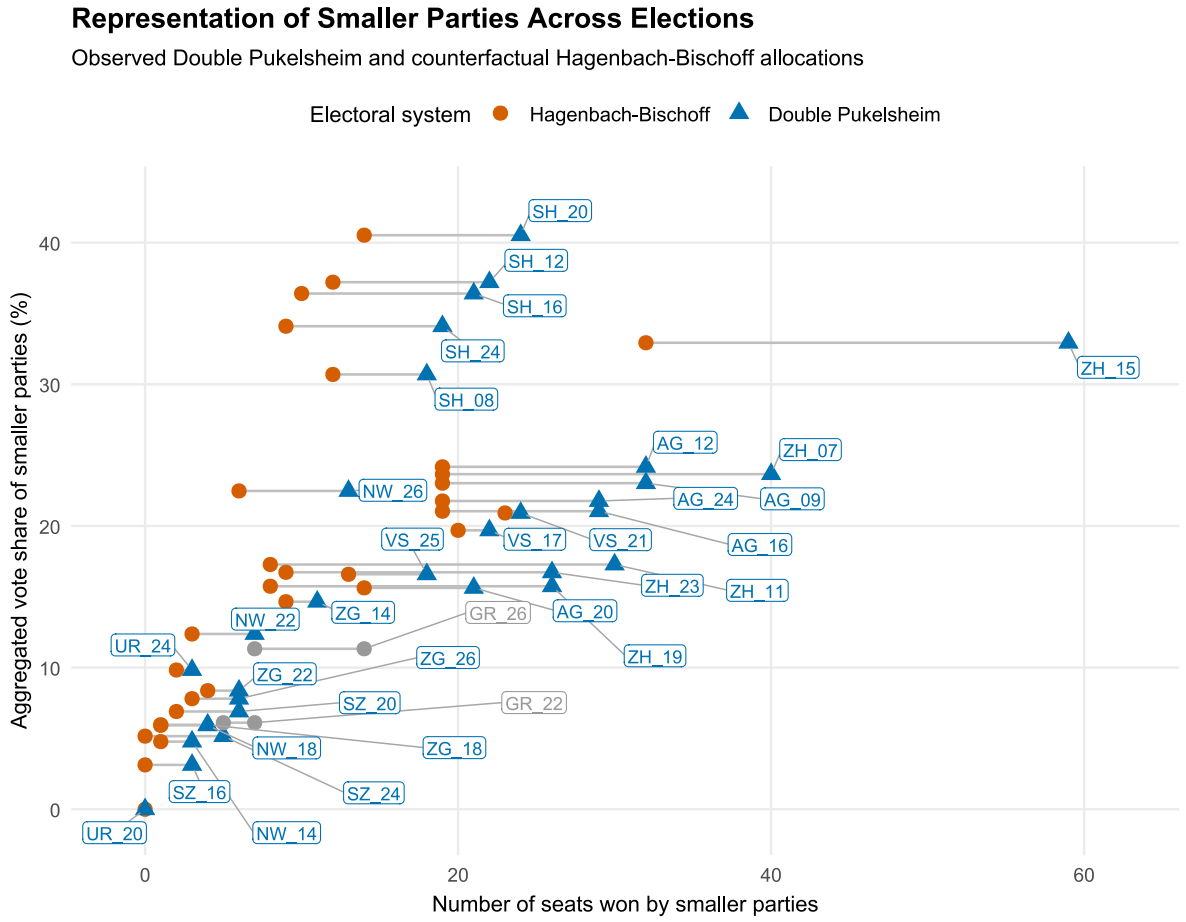


Figure 7: Smaller-party absolute seat gain across all 31 elections

10.3.2 Mechanical seat-share gains by exact party vote share

Figure 8 supplements the Figure 2 in the Analysis of Hypothesis 1. Conversely to Figure 2 it uses no grouping and plots each party-election observation at its exact vote share. It shows substantial variation in mechanical seat-share gains among parties with similar vote shares aswell as no clear monotonic relationship. Additionally to Figure 2 it shows that lists with very low vote shares cluster around zero. The Figure 8 therefore provides additional context for the group averages reported in

the main analysis. As in the grouped figure, grey open circles show Graubünden's hypothetical HB comparison for context only and the observations do not enter the transition summaries.

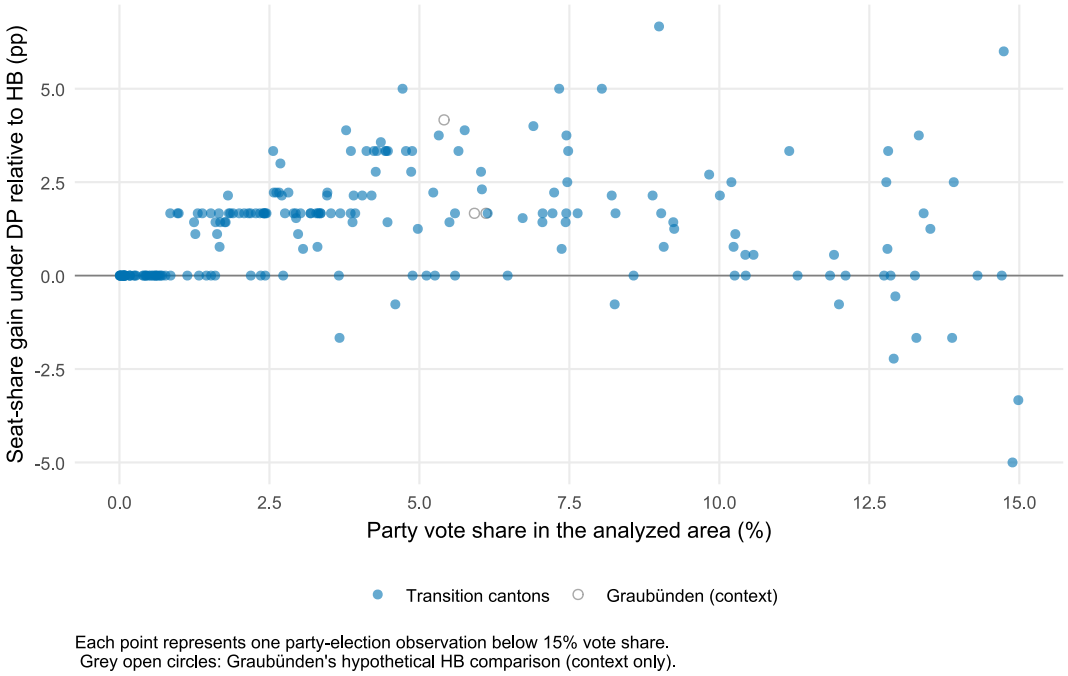


Figure 8: Mechanical seat-share gains by exact party vote share

10.4 Appendix 4: Leave-one-canton-out robustness checks

The leave-one-canton-out check, is a sensitivity and robustness check, where the statistical results of the thesis are recalculated after the omission of each single canton. Through that it can be observed, whether the results are sensitive to the inclusion of a specific canton or not. The bold values in the following tables, identify the canton omission that produces the largest absolute change from the full-sample result in each outcome column. For EQ4, this comparison is made separately within each election number. The comparisons use unrounded results.

10.4.1 Hypothesis 1

Canton omitted	Elections remaining	Mean gain (pp)	Weighted mean gain (pp)
Full sample	31	7.57	7.82
AG	26	7.56	7.88
NW	27	7.70	7.91
SH	26	**6.01**	7.01
SZ	28	7.95	8.21
UR	29	7.99	7.97
VS	28	8.16	8.61
ZG	27	8.22	8.33
ZH	26	6.78	**6.33**

Both the unweighted and weighted mean gains are relatively consistent after the exclusion of every canton. The omission of Schaffhausen and Zurich for the unweighted respectively the weighted mean.

In both cases the gain declines when they are excluded. This is aligned with the observation, that both Zurich and Schaffhausen receive relatively high seat-share gains. Further are both of them included in the observation with the maximumly possible five elections. The canton of Zurich further has the largest parliament of all analyzed cantons, giving it extra influence in the weighted mean analysis. The positive aggregate finding reported in Section 6.5.1 therefore remains robust to the exclusion of any single canton.

10.4.2 Exploratory Question 1

Canton omitted	Mean size	Median size	Share below 9	CV
Full sample	−3.06	−3.06	−1.70	−1.74
AG	−3.09	−3.09	−2.34	−2.16
NW	−4.10	−4.10	−1.94	−1.94
SH	**-1.06**	**-1.06**	**1.06**	**2.30**
SZ	−2.77	−2.77	−2.83	−3.53
UR	−4.64	−4.64	−0.27	−0.61
VS	−4.47	−4.47	−0.50	−0.78
ZG	−2.33	−2.33	−3.12	−3.97
ZH	−2.06	−2.06	−3.71	−3.19

As for Hypothesis 1, the omission of Schaffhausen comes with the highest absolute differences in all four measures. The differences for mean and median district magnitude remain negative, whereas those for the share of districts below nine seats and the CV change from negative to positive. The direction of the two latter comparisons therefore depends on the inclusion of Schaffhausen. These robustness checks therefore demonstrate sensitivity to the sample composition.

10.4.3 Exploratory Question 3

Canton omitted	Without quorum (pp)	With quorum (pp)	Difference (pp)
Full sample	9.75	6.19	3.56
AG	9.79	5.93	3.86
NW	11.29	NA	5.10
SH	**5.52**	NA	**-0.67**
SZ	NA	6.60	3.15
UR	11.43	NA	5.24
VS	NA	6.96	2.79
ZG	NA	7.00	2.75
ZH	NA	**4.23**	5.52

As in the prior robustness checks are the cantons of Zurich and Schaffhausen responsible for the highest absolute differences. The omission of Schaffhausen reverses the group difference. This leads to that without the Schaffhausen elections, the average seat-gain share is higher for elections with quorums compared to the ones without quroums. The sample for the without quroum group consists of

twelve elections, from which five are from Schaffhausen. Further do all Schaffhausen elections, experience an above average seat-share gain. Thus is the observed difference sensitive to Schaffhausen's inclusion and should not be interpreted as a stable relationship betwenn an explicit electoral quroum and the smaller party seat-share gains.

10.4.4 Exploratory Question 4

Election number	Canton omitted	Cantons	Mean gain (pp)	Mean change (pp)
1	Full sample	8	5.17	NA
1	AG	7	4.58	NA
1	NW	7	5.43	NA
1	SH	7	4.47	NA
1	SZ	7	5.47	NA
1	UR	7	5.90	NA
1	VS	7	5.68	NA
1	ZG	7	5.55	NA
1	ZH	7	**4.24**	NA
2	Full sample	8	6.80	1.63
2	AG	7	6.44	1.87
2	NW	7	7.06	1.63
2	SH	7	**5.39**	**0.92**
2	SZ	7	7.20	1.72
2	UR	7	7.38	1.48
2	VS	7	7.66	1.98
2	ZG	7	7.24	1.69
2	ZH	7	6.02	1.79
3	Full sample	7	8.36	0.97
3	AG	6	8.56	**1.49**
3	NW	6	8.64	0.85
3	SH	6	**6.69**	0.85
3	SZ	6	8.91	0.97
3	VS	6	9.11	0.62
3	ZG	6	9.33	1.34
3	ZH	6	7.25	0.67
4	Full sample	5	9.42	−0.51
4	AG	4	10.52	−0.10
4	NW	4	8.85	**-1.89**

Election number	Canton omitted	Cantons	Mean gain (pp)	Mean change (pp)
4	SH	4	**7.60**	−0.22
4	ZG	4	10.83	−0.95
4	ZH	4	9.27	0.61
5	Full sample	3	11.08	0.53
5	AG	2	13.06	**-0.28**
5	SH	2	**8.29**	0.79
5	ZH	2	11.90	1.07

For exploratory question four, the robustness check was done for each election number separately. Zurich or Schaffhausen produces the larges absolute change in six out nine comparisons. The mean seat-share gains remain positive under every omission. However, the mean change from election three to four switches from negative to positive when Zurich is excluded, while the change from election four to five switches from positive to negative when Aargau is excluded. These later transitions include only five respectively three cantons. This makes their averages particularly sensitive to individual cantons.

10.5 Declaration of AI-Assistance

10.5.1 Self Reflection on AI-Usage

−> add how to deal with tendency of LLMs to glaze you −> e.g. say reason why you think one should consider this particular aspect or not, then respond to me with a suggestion −> add how to deal with LLMs giving too long texts −> repetitive explanations −> add AI better than me at structuring, table of contents, finding discrepancies there etc